1 Introduction

1.1 Cellular Heterogeneity and Transcriptomic Profiling

The human immune system operates as a heterogeneous population of specialized cellular phenotypes. Peripheral Blood Mononuclear Cells (PBMCs) provide a direct, accessible window into this circulating network, comprising lymphocytes (T cells, B cells, NK cells) and myeloid lineages (monocytes, dendritic cells). Historically, the resolution of these populations relied on restricted surface-protein markers. Single-cell RNA-sequencing (scRNA-seq) permits the quantification of the entire transcriptome within individual droplets.

scRNA-seq generates a gene expression count matrix. Within this high-dimensional data structure, distinct cell types group into observable clusters. Downstream analysis groups cells into clusters.

1.2 Limitations of Standard Clustering Methodologies

While the scRNA-seq provides high-resolution transcriptomic data, the computational methodologies used to resolve the resulting matrices remain subject to technical limitations. Single-cell data is subject to technical noise: ambient RNA leakage from lysed cells, stochastic droplet multiplets (doublets), and extreme variance in sequencing capture efficiency.

Rather than addressing this noise with standard statistical models, conventional downstream pipelines frequently operate as unsupervised models requiring arbitrary parameterization. They rely on arbitrary parameter selection—such as guessing the number of nearest neighbors (k) or community resolution (r)—until the output visually aligns with human expectation. Iterative parameter adjustment introduces subjective bias.

Standard workflows are susceptible to the following statistical errors:

- **Data Leakage:** Computing global variance (highly variable genes) and dimensional reduction (PCA) on the entire dataset simultaneously, thereby allowing the training set to incorporate data from the validation set.
- **Double Dipping (Circular Inference):** Utilizing the exact same cellular vectors to dynamically define cluster boundaries and subsequently compute the statistical significance of the marker genes defining those boundaries. This circular inference inflates statistical significance, increasing the false-discovery rate for marker genes.

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2 Methods

2.1 Pre-Phase 1: Upstream Provenance and System reads Boundaries

2.1.1 Data Provenance and Upstream Processing

PBMCs were isolated using density gradient centrifugation. Single-cell encapsulation and barcoding were executed utilizing the 10x Genomics Chromium droplet-based microfluidic platform. Within the generated Gel Bead-in-emulsions (GEMs), cellular lysis and bead dissolution facilitated the capture of polyadenylated mRNA via Poly(dT) primers linked to unique Cell Barcodes and Unique Molecular Identifiers (UMIs). Following reverse transcription and Template Switch Oligo (TSO)-mediated extension, the resulting complementary DNA (cDNA) libraries were amplified and sequenced via Illumina Sequencing by Synthesis (SBS).

Primary upstream analysis was conducted by transposing binary BCL outputs into FASTQ format. Alignment to the GRCh38/hg19 human reference genome was performed utilizing the STAR splice-aware alignment algorithm, producing structurally mapped BAM files. Subsequent feature quantification intersected genomic coordinates with the corresponding Gene Transfer Format (GTF) annotation, collapsing redundant UMIs to eliminate exponential amplification bias. This generated the raw transcriptomic count matrices, structured natively in a sparse Coordinate List (COO) format with a strict `Cells x Genes` orientation.

2.1.2 Local Hardware Constraints and Decontamination Bottlenecks

Standard methodologies for the computational removal of ambient RNA contamination (e.g., CellBender) rely heavily on deep convolutional neural networks and generative modeling. These architectures are optimized for execution on discrete NVIDIA CUDA tensor cores. Local computation was executed on an ARM64 Apple Silicon framework, which leverages unified Metal Performance Shaders (MPS). Compiling and executing CUDA-dependent epoch-training loops natively on MPS introduces significant memory bottlenecks and extended processing times, extending execution times for generative ambient RNA modeling.

2.1.3 The Structural reads Boundary

To accommodate these hardware constraints, this pipeline initiates downstream of deep-learning-based ambient RNA decontamination methods. Analysis was initiated utilizing the 10x Genomics Filtered Feature-Barcode Matrix, which has undergone heuristic, rank-based UMI thresholding (the conventional "Knee Plot" inflection) to isolate high-confidence cellular barcodes from ambient background partitions.

To account for residual ambient RNA, the downstream Phase I Quality Control steps employ robust variance filters. Median Absolute Deviation (MAD) was applied as a robust, non-parametric distance metric, deploying an absolute 5-MAD boundary to isolate structural anomalies such as mitochondrial stress events, high ribosomal gene expression, and doublets. This filtering step establishes the thresholds prior to normalization.

The physical sample consists of PBMCs extracted from a healthy human male donor, aged 18-35 were obtained by 10x Genomics from Cellular Technologies Limited.

2.1.4 Sequencing Architecture and Chemistry

Single-cell capture and library preparation were performed using the Gene Expression libraries which were generated as described in the Chromium GEM-X Single Cell 3' Reagent Kits v4 User Guide (CG000731) and sequenced on an Illumina NovaSeq 6000 with approximately 39,000 read pairs per cell for the Gene Expression library.

Paired-end, dual indexing libraries were sequenced with this configuration:

- 28 cycles Read 1
- 10 cycles i7
- 10 cycles i5
- 90 cycles Read 2

Libraries were analyzed using the `cellranger multi` pipeline.

2.1.5 Reference Genome Alignment

The translation of raw sequencing reads into the count matrix $X \in \mathbb{R}^{C \times G}$ was performed utilizing the Cell Ranger pipeline (version 9.0.0). The reads were aligned against the GRCh38 human reference genome. Consistent use of this reference genome is critical, as utilizing a different genomic assembly would alter the gene feature space and hinder downstream reproducibility. The resulting filtered feature-barcode matrices (`matrix.mtx` , `barcodes.tsv` , `genes.tsv`) serve as the primary input for the downstream analytical pipeline.

2.2 Phase 1: Transcriptomic Quality Control and Matrix Preprocessing

2.2.1 Matrix Initialization and Quality Control Metrics

Initial quality control establishes the viability of the sequenced droplets. Upon ingesting the Coordinate List (COO) matrix, the pipeline computes fundamental quality control metrics—referred to as basic quality control metrics. For each cellular vector, the pipeline calculates the total library size (total_counts) and the number of detected features (n_genes_by_counts).
The proportion of reads mapping to mitochondrial and ribosomal genomes was calculated. These fractions serve as primary proxies for cellular stress and active apoptosis. During the microfluidic encapsulation process, a mechanically ruptured cell membrane will leak diffuse cytoplasmic mRNA into the ambient medium, whilst the intact mitochondria remain trapped within the oil partition. Consequently, an artificially inflated mitochondrial read fraction (`pct_counts_mt`) indicates a compromised or non-viable cell that must be excluded prior to downstream modeling.

2.2.2 Limitations of Parametric Thresholding and Stochastic Multiplets

Conventional data science pipelines frequently employ Standard Deviation (SD) to define threshold boundaries. However, SD assumes the underlying data follows a Gaussian (normal) distribution.

Single-cell RNA sequencing data is non-Gaussian; it represents a highly skewed, continuous biological gradient. Employing a mean-dependent metric like SD on skewed biological tissue allows extreme outliers (such as ambient RNA pools or cell clumps) to disproportionately skew the mean. This artificially expands the threshold bounds, resulting in the erroneous retention of dead cells and debris.

Furthermore, droplet encapsulation is a stochastic Poisson process. There is a non-zero probability that two distinct cells are co-encapsulated within a single Gel Bead-in-emulsion (GEM). These "doublets" exhibit artificially inflated complexity vectors and hybrid transcriptomic profiles that, if undetected, generate spurious continuous differentiation trajectories where none biologically exist.

2.2.3 5-MAD QC Filtering and Doublet Removal

Data filtering using MAD and doublet removal.

Quality control filtering used Median Absolute Deviation (MAD). The pipeline implements a strict non-parametric outlier identification algorithm utilizing Median Absolute Deviation (MAD). By calculating absolute distances from the spatial median rather than the mean, the MAD metric remains highly resistant to the skewing effects of extreme outliers. The pipeline applies an absolute **5-MAD threshold**, dynamically establishing highly restrictive variance bounds for `total_counts`, `n_genes_by_counts`, and `pct_counts_mt`. Cells exceeding a 5-MAD threshold were removed. Features (genes) detected in fewer than three viable cells are simultaneously dropped to eliminate extreme sparsity and reduce tensor dimensionality.

To identify potential multiplets, we utilized Scrublet, which simulates synthetic doublets by sampling and combining observed transcriptomic profiles to calculate a doublet density score for each barcode. High-scoring multiplets are subsequently removed from the matrix.

Finally, to prepare the tensor for topological extraction while preserving the unedited baseline data, A multi-layer framework was established. The raw, discrete integer counts are stored in a base `counts` layer. The copy of primary `.X` tensor is then subjected to a depth-normalization algorithm (scaling total counts to a target sum of 10^4 to correct for stochastic sequencing depth) and a log-plus-one (`log1p`) transformation to stabilize variance. This transformed data is stored in the `log1p_norm` layer of the primary anndata. The fully decontaminated, variance-stabilized matrix is then saved to disk, ensuring the subsequent dimensionality reduction pipeline operates exclusively on high-confidence, viable cellular profiles.

2.3 Phase 2: Dimensionality Reduction and Subpopulation Clustering

This phase separates biological phenotypes. Operating on the quality-controlled master tensor $X \in \mathbb{R}^{C \times G}$, where the row space C represents viable cellular droplets and the column space G represents genomic features, Cells were grouped into subpopulations. to ensure the defined clusters represent robust biological states rather than algorithm-specific artifacts, this pipeline applies a strict physical segregation of data, followed by iterative variance stabilization, hyperparameter optimization, and out-of-sample validation.

2.3.1 Dataset Segregation and the Prevention of Data Leakage

If a dataset is utilized in its entirety to establish the clustering boundaries which are inherently biased by the specific topological noise of that exact dataset. In the standard analysis of single-cell transcriptomics, this principle is frequently violated by two distinct, yet linked, statistical errors: **Data Leakage** and **Double Dipping**.

Data Leakageoccurs when global variance structures are computed prior to data splitting. It occurs when global variance structures—such as highly variable genes (HVGs) or the eigenvectors of a principal component analysis (PCA)—are computed on the entire dataset. If the training manifold is built upon global variance, the geometric coordinates of the validation set have already been incorporated into the training state, rendering independent validation impossible.

Double Dipping is a circular inference error. It occurs when an algorithm uses a specific dataset to define physical cluster boundaries, and then subsequently uses that exact same data to run differential gene expression (DGE) tests between those boundaries. Because the transcripts driving the clustering are inherently the ones being tested for statistical significance, this results in false-positive marker genes, yielding false-positive marker genes and driving spurious cluster formation.

The data was split into training and testing sets to prevent data leakage. Upon loading the standardized, quality-controlled expression matrix, the cellular barcodes are bisected into two independent, non-overlapping subsets: X_{train} and $X_{project}$. This separation is executed using a stratified random split (allocating 50% of the data to each tensor) locked by a global pseudo-random seed to guarantee exact computational reproducibility. All subsequent identification of highly variable features, calculation of dimensional manifolds, and definitions of physical cluster boundaries are restricted entirely to the X_{train} tensor. The $X_{project}$ matrix is withheld from analysis until the final validation phase

2.3.2 Statistical Methods for Dimensionality Reduction and Clustering

Each module is engineered to mitigate specific forms of technical noise, substituting heuristic approaches with robust quantitative frameworks.

I. Cell Cycle Confounder Evaluation

Cellular division is a significant driver of transcriptional variance. Cell cycle variance can confound clustering outputs based purely on whether the cells are in the resting (G0/G1), DNA synthesis (S), or mitotic (G2/M) phases. S-phase and G2M-phase gene scores were calculated to control for cell cycle effects. By projecting these scores onto the foundational cluster, we visually and quantitatively audit the manifold. If the primary topological axes align precisely with cell cycle progression rather than immune phenotype, mathematical regression of these vectors is mandated.

II. Analytic Variance Stabilization (NPR, HVG, and PCA Recalibration)

Raw transcript counts are heavily confounded by the arbitrary total sequencing depth of each individual droplet. Traditional logarithmic normalization often fails to stabilize variance in highly sparse datasets, artificially inflating the weight of lowly expressed, noisy transcripts. To isolate the true biological signal, Analytic Pearson Residuals were calculated. This transformation models the expected expression of a transcript based on the physical sequencing depth of the cell and the global relative abundance of the gene across the population. The residual $r_{i,j}$ for cell i and gene j is calculated as:

$$r_{i,j} = \frac{x_{i,j} - \mu_{i,j}}{\sqrt{\mu_{i,j} + \frac{\mu_{i,j}^2}{\theta}}}$$

Here, $x_{i,j}$ is the observed count, $\mu_{i,j}$ is the expected count under a null model of uniform distribution, and θ is the overdispersion parameter. In single-cell RNA-sequencing, biological variance scales quadratically with the mean expression. Rather than estimating this arbitrarily, θ

is locked to 100, adhering to the empirical consensus established by the Seurat `SCTransform` framework (Hafemeister & Satija). This precise threshold prevents ultra-abundant transcripts from exhibiting infinite variance and distorting the Euclidean space.

Furthermore, to ensure the resulting low-dimensional geometry maps functional biological states rather than cell death, transcripts corresponding to mitochondrial and ribosomal structures are algorithmically excluded from the Highly Variable Gene pool. While apoptosis generates a high variance signal, it represents a dying state, not a stable immune phenotype. Dimensionality reduction was performed via principal component analysis (PCA) on the highly variable genes, explicitly excluding mitochondrial and ribosomal transcripts.

III. Homogenous Cluster Assessment (Automated Over-Clustering Prevention)

Recursive clustering algorithms are susceptible to over-clustering. We must define the boundary between a true biological sub-population and an amorphous cloud of technical variance.

In principal component analysis (PCA), technical variance generates variance eigenvalues up to a theoretical limit described by the Marchenko-Pastur distribution. An eigenvalue that is only 1.8 or 2.0 times larger than the baseline noise often represents Tracy-Widom fluctuations. To identify populations, the Structural Energy Ratio was calculated: the variance of the primary axis of separation (PC1) divided by the median baseline thermal noise of the matrix. We enforce a strict 5-sigma eigenvalue threshold of 3.5. If the structural ratio is ≥ 3.5, an angular structural divergence (an "elbow") exists, confirming a true sub-population. If the ratio falls below 3.5, the geometry is classified as isotropic, and the module identifies the sub-population as an homogenous sub-population, halting further algorithmic subdivision.

IV. Hyperparameter Optimization for Cluster Stability

Graph-based community detection algorithms (like Leiden) rely heavily on two highly sensitive parameters: the scaffolding limit (k-nearest neighbors) and the leiden resolution (r). The standard industry practice of selecting these scalars via iterative "hit and trial" introduces severe human bias. To optimize parameters, a grid search was executed to map modularity.

To correct this error, parameter stability was mapped by computing the structural Modularity (Q) across an extensive Cartesian grid of k and r coordinates. This generates a matrix of resulting cluster counts. Instead of guessing, A grid search was executed across the parameter space for an optimal parameter coordinate region (e.g., 2×2 or larger) where the number of discrete clusters remains entirely static despite aggressive perturbations in both k and r. By generating contour and heat maps of this surface, we identify the geometric median of this stable parameter region. This narrowed the parameter selection to 2 or 3 options.

V. Cluster Stability Bootstrapping (Jaccard Overlap)

Even optimized boundaries must survive iterative data subsampling to be considered biologically real. Upon generating a baseline K-Nearest Neighbors graph, the pipeline executes a rigorous bootstrap diagnostic. 20% of the cells are randomly removed, and the community detection process is recalculated from scratch across multiple iterations. A Jaccard Overlap score is calculated between the original boundary and the newly generated boundary. We enforce that a mean Jaccard score > 0.85 indicates high topological stability, while scores between 0.62 and 0.85 indicate moderate stability requiring careful validation during annotation.

2.3.3 Global Lineage Resolution (Hyperparameter grid search)

With the computational methods defined, the active execution of the X_{train} tensor begins. The objective of the Hyperparameter grid search is to resolve the primary, overarching lineages within the PBMC population (e.g., distinguishing the entire T-cell compartment from the Myeloid compartment).

The global matrix undergoes variance stabilization via Analytic Pearson residuals, and the top highly variable genes are isolated to construct the primary PCA manifold. A hyperparameter grid search evaluates a broad parameter space ($k \in 5, 105$, $r \in 0.01, 0.21$) to lock the optimal macro-resolution. Once these primary boundaries are verified by the Jaccard bootstrapping diagnostic, the global matrix is partitioned. The X_{train} matrix was subset into independent matrices, each representing a single, distinct macro-cluster, and saves these isolated matrices to disk for independent processing.

2.3.4 Recursive Subclustering

The variance required to differentiate subtle micro-phenotypes (e.g., a naive CD4+ T-cell versus a memory CD4+ T-cell) is independent of and smaller than the variance that separates major macro-lineages. In a global PCA, this critical micro-variance is unrepresented in the primary principal components.

Therefore, each isolated macro-cluster matrix is ingested individually to reconstruct its internal geometry from the ground up. To prevent over-clustering the data, this pipeline bounds the recursion depth to a two-tier architecture: one Macro-level separation, followed by exactly one Micro-level separation. The global highly variable genes are discarded. The Analytic Pearson Residuals are recalculated specifically on the isolated sub-population, exposing a new, highly specialized subset of local HVGs. A new, localized PCA space is computed.

Before further subdivision occurs, the Clustering Convergence is executed. Clusters with a Structural Energy Ratio < 3.5 were classified as homogenous and excluded from further sub-clustering. If the ratio is ≥ 3.5, a high-resolution grid search optimization for Leiden modularity is executed to find the exact boundaries of the sub-populations. This loop recurs until every physical compartment of the data reaches an isotropic variance state, ensuring comprehensive variance partitioning.

2.3.5 The Out-of-Sample Projection Audit

The final phase evaluates the generalizability of the learned topological boundaries. The $X_{project}$ matrix, withheld during the training phase, is introduced.

A standard, critical flaw in validation pipelines is allowing the test set to normalize itself. If $X_{project}$ is permitted to calculate its own internal mean and variance for normalization, the testing space has influenced itself, resulting in data leakage. To validate the clustering model, $X_{project}$ is excluded from defining its own parameters. It is normalized using the global gene probabilities (p_j) derived from the X_{train} tensor, multiplied by its own local cell depths to calculate the expected variance.

These rigidly normalized validation vectors are then projected onto the pre-computed PCA eigenvectors of the training space. Using the finalized K-Nearest Neighbor architecture, the Cells in the validation set were classified based on expression profile similarity to the training set.. Classification of the testing set using the training model demonstrated robust quantitative generalizability.

2.4 Phase 3: Differential Marker Identification and Lineage Validation

Following the mathematical resolution of topological boundaries, the system must assign biological identities to the identified clusters. This phase dictates the extraction of defining genomic features (marker genes) for each isolated cluster. To ensure statistical significance, the pipeline utilizes a combination of statistical filtering, adaptive thresholding, and negative marker assessment to ensure cluster purity.

2.4.1 Statistical Framework for Feature Identification

Before performing the extraction across the data hierarchy, we establish the specific computational methods designed to isolate the true biological signal from background transcriptional noise.

I. Statistical Filtering and Cluster Size Constraints

Cellular phenotypes are generally defined by is that a cellular state is defined by transcripts that are differentially upregulated relative to other clusters. However, executing statistical tests on clusters with low cell counts increases the likelihood of capturing technical noise. To resolve this, A size-based filter was applied: any topological state containing fewer than 10 cellular vectors is excluded from differential expression analysis to avoid statistical bias associated with low-n clusters. For the remaining viable states, a Wilcoxon Rank-Sum test is executed across the log-normalized layer. To prevent the capture of biologically irrelevant noise, the resulting output is filtered using standardized thresholds: only genes exhibiting an adjusted p-value < 0.05 and a strict $\text{Log}_2\text{FoldChange} > 1.0$ (representing a minimum two-fold absolute biological up-regulation) are permitted to proceed.

II. Adaptive Thresholding for differential gene expression (DGE)

Standard differential gene expression (DGE) tests often return thousands of statistically significant genes for large macro-clusters, complicating the identification of primary lineage-defining markers. Conversely, highly specific micro-lineages may return almost zero genes under universal thresholds. An adaptive p-value threshold was applied.

For robust clusters, the system isolates only the genes existing in the 93.75th percentile ($Q_{93.75}$) of statistical significance (calculated via negative log-adjusted p-values). However, if a cluster yields fewer than 3 significant genes under the Q93.75 cut, The threshold is programmatically reduced to extract exactly 3 markers. Once distilled, these markers are sorted not just by magnitude, but by their *Violin Delta*—the mathematical difference between the percentage of cells expressing the gene in the target cluster versus the background reference. This guarantees the selection of highly specific, cluster-defining markers (highly prevalent in the target, absent elsewhere) rather than universally expressed genes that simply experienced a slight up-regulation.

III. Negative Marker Assessment and Purity Validation

Annotation relying solely on positive identification can mask doublet contamination. To enforce Negative Marker Validation, we recognize that identifying T-cell markers in a cluster does not prove it is a pure T-cell population; it could be a doublet population expressing macrophage markers. A true biological cell is defined just as much by the genes it actively downregulates as by the ones it expresses.

To assess cluster purity, the pipeline evaluates negative markers characteristic of non-target lineages. It systematically assesses markers from alternative lineages to verify transcriptomic purity and the absence of cross-contamination. If other lineage markers manifest strongly within the target state, topological contamination is flagged.

IV. Algorithmic Reference Mapping

Manual annotation can be influenced by the subjective interpretation of existing literature. To establish an unbiased empirical baseline, the system incorporates CellTypist, an automated supervised machine learning framework. Using pre-trained, high-fidelity immune models (`Immune_All_High` and `Immune_All_Low`), the system executes a majority-voting algorithm across the Leiden boundaries. This provides an algorithmic hypothesis for every cluster, providing an independent reference to validate the statistically derived markers.

2.4.2 Implementation of the Marker Identification Pipeline (Phase III Execution)

The script processes the cluster metadata generated during Phase II, ensuring data traceability. The analysis is performed hierarchically as follows:

1. **Macro-State Marker Identification:** The X_{train} matrix was analyzed, and its anchored macro-boundaries are subjected to the Wilcoxon extraction pipeline. Clusters below the minimum cell threshold are removed, adaptive thresholds are applied, and the top markers are isolated and rendered. The global structure is concurrently mapped via the automated CellTypist `Immune_All_High` model.
2. **Micro-State Resolution:** The script processes the physical file paths of every isolated micro-tensor. For each validated micro-state, the extraction process is re-initialized within its local geometric space, mapping the high-resolution features against the CellTypist `Immune_All_Low` model. If a subset was previously locked as an isotropic Terminal State in Phase II, extraction is correctly bypassed, and it inherits the parent topology's markers.
3. **Cross-Validation and Wide-Span Auditing:** Every extracted micro-state is systematically subjected to the Negative Marker Cross-Validation to prove spatial purity. Concurrently, the states are validated against a pre-curated canonical JSON ledger of established immune markers to align the empirical findings with accepted biological literature.
4. **Metadata Ledger Generation:** The final step involves generating standardized JSON templates to facilitate the manual assignment of biological labels and Cell Ontology IDs (`annotation_manual.json` and `ontology_cl_id_manual.json`). These ledgers map the exact mathematical boundaries discovered in the data, formatting the environment for human annotation and linking the annotations to the coordinate data.

2.5 Phase 4: Metadata Annotation, Ontological Alignment, and Dataset Integration

The final phase of the analytical pipeline bridges the gap between cluster coordinates and biological identities. It is at this stage that the rigorous quantitative evidence generated in previous phases is synthesized into cellular identities. Metadata was integrated to generate the final dataset.

2.5.1 Framework for Metadata Integration

Before detailing the orchestration, the specific mechanical safeguards enforcing annotation validity must be defined.

I. Independent Metadata Mapping via JSON Ledgers

Biological labels were recorded systematically to ensure reproducibility.

To resolve this, the system separates the metadata mapping from the primary data processing. Metadata was mapped using `annotation_manual.json` and `ontology_cl_id_manual.json`. These are the exact structured, but functionally empty, taxonomic ledgers generated during the terminal step of the previous marker extraction phase. They ensure consistent data traceability. The system maps the string values populated within these ledgers to the numeric Leiden keys of the isolated physical matrices. Furthermore, to satisfy rigorous Interoperability, the system applies the mapping of standardized Cell Ontology (CL) IDs alongside colloquial cell names, ensuring the resulting data can be ingested by any global database without semantic ambiguity.

II. Hierarchical Annotation Inheritance

The iterative subsetting of the matrix generated a nested structure of both active states (micro-lineages undergoing active clustering) and locked states (exhibiting isotropic variance). The annotation pipeline must navigate this spatial hierarchy seamlessly. For active states, labels are mapped directly via their localized topological keys. However, for clusters previously locked as isotropic Variance States, the local key is non-existent. Terminal clusters were identified and assign labels based on parent-level lineage definitions. Lineage was traced through the DAG, and the macro-level phenotype was assigned., ensuring all cells are correctly assigned within the hierarchical structure.

2.5.2 Data Integration and Final Dataset Generation (Phase IV Execution)

The execution of the annotation pipeline operates in a three-stage protocol designed to prevent barcode duplication and ensure consistent label assignment.

1. **Training Set Integration and Ledger Assembly (The X_{train} Matrix):** The system loads the state dictionary for the primary training matrix (X_{train}). It iterates through every isolated macro and micro tensor currently on disk. For each physical file, the pipeline parses the human-annotated JSON dictionaries and maps the corresponding biological identities and CL IDs. Once localized mapping is complete, the pipeline generates a mapping of cell barcodes to biological identities and CL IDs `cell_barcode, manual_label, human_CL_ID` from each sub-matrix. These vectors are aggregated and concatenated into a singular, central CSV ledger (`master_labels_df.csv`).
2. **Validation Set Integration (The $X_{project}$ Matrix):** The pipeline then loads the state dictionary for the out-of-sample projection matrix ($X_{project}$). It repeats the identical

localized injection process. When appending the resulting validation vectors to the central CSV ledger, the system performs a deduplication check. If a barcode collision is detected (a highly improbable event under the current architecture, but a mandatory CRISP-DM robustness safeguard), the system isolates the conflict and prioritizes the most recent RAM state, finalizing the metadata of cell identities.

3. **Global Dataset Integration:** The final step generates the output matrix. The pipeline loads the raw, un-split, quality-controlled global matrix and the finalized central CSV ledger. Utilizing a highly optimized, vectorized `.map()` operation, the biological identities and CL IDs are projected from the CSV ledger directly onto the coordinates of the global tensor. Cells without labels (due to prior quality control filtering) are assigned a status of 'Unknown/Filtered'. The fully integrated, universally labeled object is then serialized to disk as the final ML-Ready artifact (`_ML_Ready.h5ad`), completing the analytical pipeline.

2.6 Streamlit API: Development of an Interactive Interface for Human-in-the-Loop Analysis

To facilitate manual review of the clustering outputs, In high-resolution bioinformatics, automated pipelines can be limited by their inability to resolve biological edge cases during the deployment phase. To resolve this, the system culminates in a graphical user interface (built upon the Streamlit framework). An interface was utilized for manual review of the clustering outputs.

2.6.1 The Reactive Memory Constraint and Tensor Pinning

In single-cell transcriptomics, reading a large `.h5ad` expression tensor from the physical disk takes significant time and RAM. If the application attempts to reload the matrix from disk on every reactive cycle, it risks exceeding memory capacity and causing application instability.

To address these resource constraints, the user interface utilizes two distinct memory-preservation instruments:

1. **Resource Caching (Tensor Pinning):** The `load_tensor` function was isolated from the reactive execution loop using a static cache decorator (`@st.cache_resource`). When the matrix is loaded once, it is permanently pinned in the server's RAM. Subsequent UI interactions simply reference this existing memory block rather than triggering redundant, heavy disk I/O operations.
2. **Session State Management:** Transient states—such as the human-approved k and r coordinates, or the active queue of micro-lineages waiting to be processed—are registered into a protected dictionary (`st.session_state`). This facilitates a persistent state that is maintained across subsequent UI interactions of the UI, allowing the user to progress sequentially through the Directed Acyclic Graph without losing their previous structural decisions.

2.6.2 Integrated Iterative Validation Breakpoints

Standard pipelines often output final clusters without intermediate evaluation nodes. This explicitly violates the requirement for iterative evaluation.

Execution breakpoints were utilized to permit manual review of stability metrics before finalizing hyperparameters. Instead of running Phase II (Clustering) as a single continuous script, the pipeline divides the execution into a dynamic queue.

1. **Macro-Level Optimization:** The initial parameter sweep was executed, and stability metrics were reviewed prior to finalization.
2. **The Micro-Queue:** Once the Macro-State is locked and the matrix is fractured, the resulting sub-matrices are pushed into an active Python list (the `micro_queue`). Localized stability was evaluated iteratively for each micro-state to allow manual parameter input. Execution pauses were implemented at these nodes to allow manual review.

2.6.3 Data Visualization and DOM Isolation

To allow the human analyst to make informed parameter decisions, the geometric evidence (scatter plots, dendrograms, stability surfaces) must be visualized within the interface. However, rendering mathematical Scalable Vector Graphics (SVGs) directly into a web interface can lead to layout inconsistencies.

The visualization module addresses this by standardizing the image rendering protocol. Instead of merely linking to the image, the pipeline opens the physical `.svg` file and utilizes regular expressions to modify the XML attributes, ensuring the height and width are responsive within the interface. to force `width="100%"` and `height="auto"`. The rewritten XML is encoded into a Base64 string and injected directly into a rigidly styled CSS flexbox container. This ensures that regardless of the physical dimensions of the manifold generated by the backend pipeline, the visual evidence is perfectly constrained and isolated within the browser's Document Object Model (DOM).

2.6.4 Workspace Initialization and Environment Reset

The integrity of a scientific workspace is compromised if residual data from previous experiments contaminates the current matrix. To ensure a reproducible environment, the interface includes a function to clear the workspace cache, utilizing `shutil.rmtree` to systematically remove previous physical data directories and temporary memory artifacts. It then recreates the directory structure, ensuring a consistent baseline for subsequent data processing.

Finally, the Annotation pipeline tab provides the interface for Phase IV. Instead of forcing the analyst to write code to label cells, the system parses the empty JSON taxonomies generated in Phase III and renders them as an interactive, two-dimensional dataframe. The user inputs their biological deductions and standardized Cell Ontology (CL) IDs directly into the matrix. Upon validation, the system records these labels to disk as standardized JSON files as JSON files, immediately triggering the metadata integration script to merge these human labels into the final ML-Ready tensor. This strict separation between human UI input and backend tensor mutation guarantees absolute computational provenance.

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3 Results

The downstream PBMC analysis pipeline was applied to the `5k_Human_Donor1_PBMC_3p_gem-x` dataset. The workflow was executed using automated parameter selection for dimensionality reduction and clustering. The results of the analysis are detailed below.

3.1 Data reads and Quality Control

Initial quality control (QC) and data filtering (`p03`) were performed to remove technical artifacts prior to downstream clustering. The raw transcriptomic dataset was filtered to remove technical noise and retain high-quality cells.

Initial State:

The raw sequencing data generated an initial matrix consisting of `5710` cells and `38606` features (genes).

Technical Filtering:

Standard QC metrics were used to filter out low-quality cells:

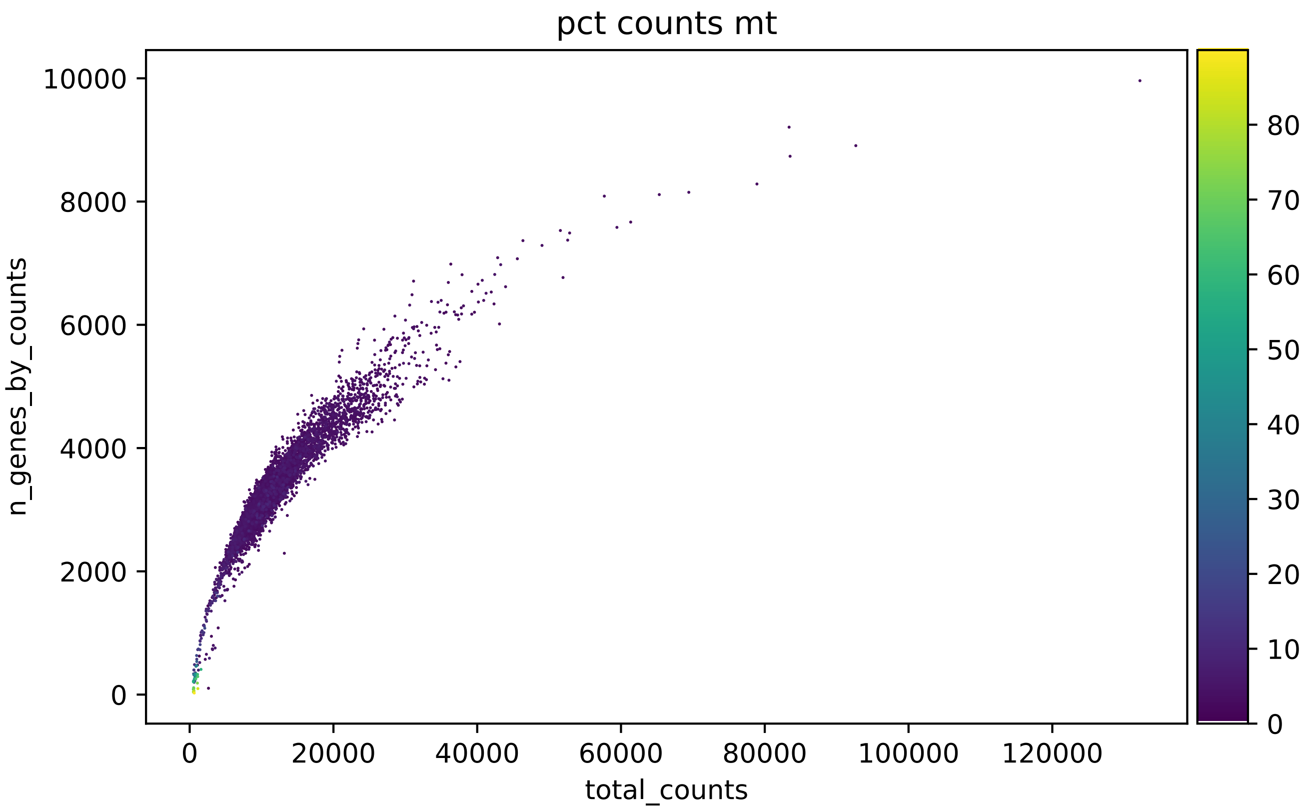
- **Library and Feature Complexity:** Cells exhibiting significant deviations in total transcript abundance or unique gene counts were filtered using a threshold of `5–MAD` (Median Absolute Deviation) and of genes whose expressions were in less than 3 cells.
- **Mitochondrial Fraction:** To exclude dead, dying, or stressed cells that contribute to ambient RNA contamination, cells with a mitochondrial transcript fraction exceeding `5–MAD` were removed.
- Removing total of 377 cells and 13741 genes

Doublet Removal:

Following initial QC, computational doublet detection (`doublet removal`) was executed to identify and remove `11` predicted multiplets. This step is critical to prevent the clustering algorithm from generating false hybrid-states between distinct biological lineages.

Final QC State and Data Split:

Following filtering and doublet removal, the final dataset comprised `5322` high-quality cells and `24865` genes. To prevent data leakage and circular inference (double dipping) during downstream marker extraction, the filtered matrix was randomly bisected (`random_split_data` , seed 42), allocating 2,661 cells to the training set (X_{train}) and 2,661 cells to the projection holdout set ($X_{project}$).



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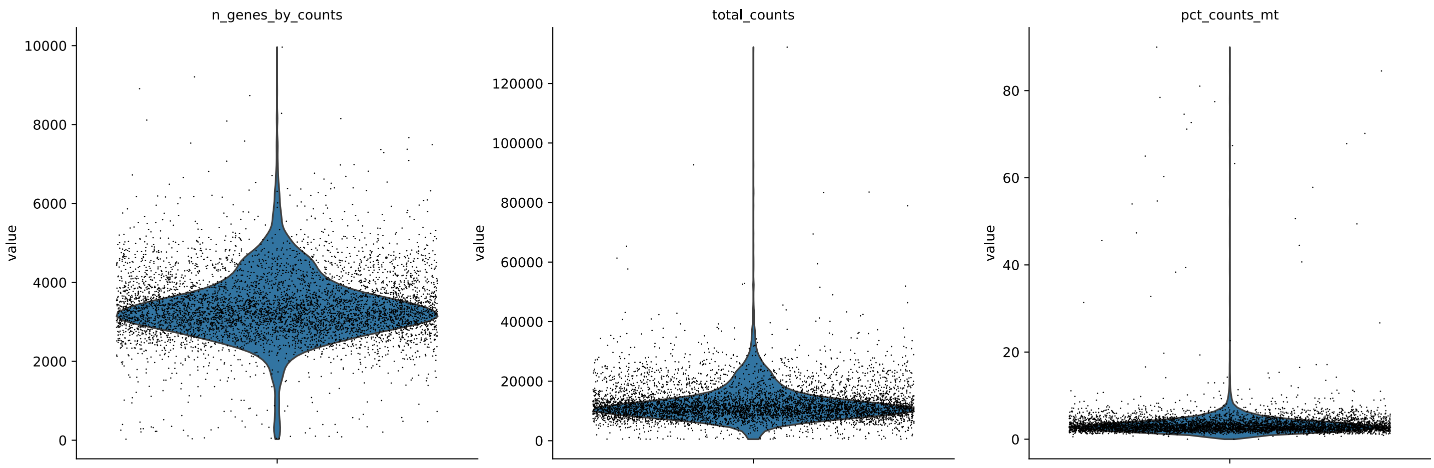
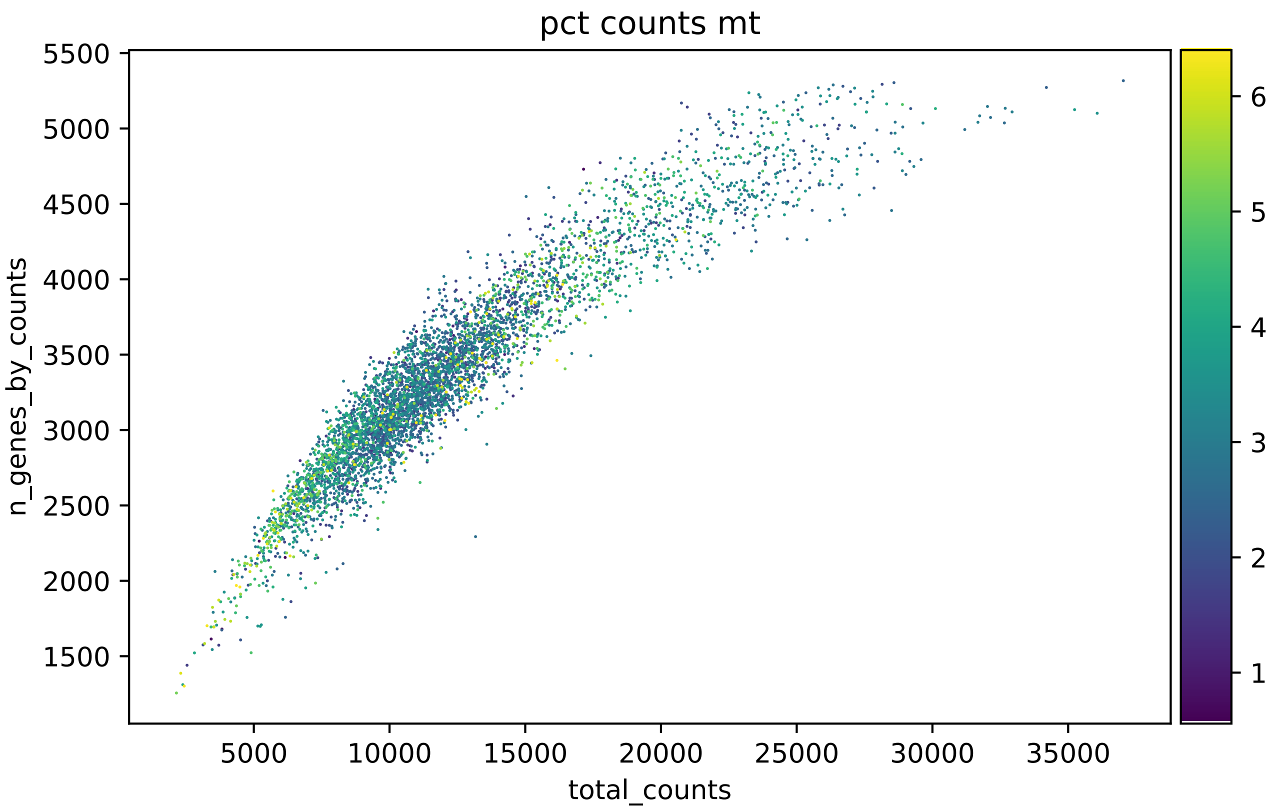
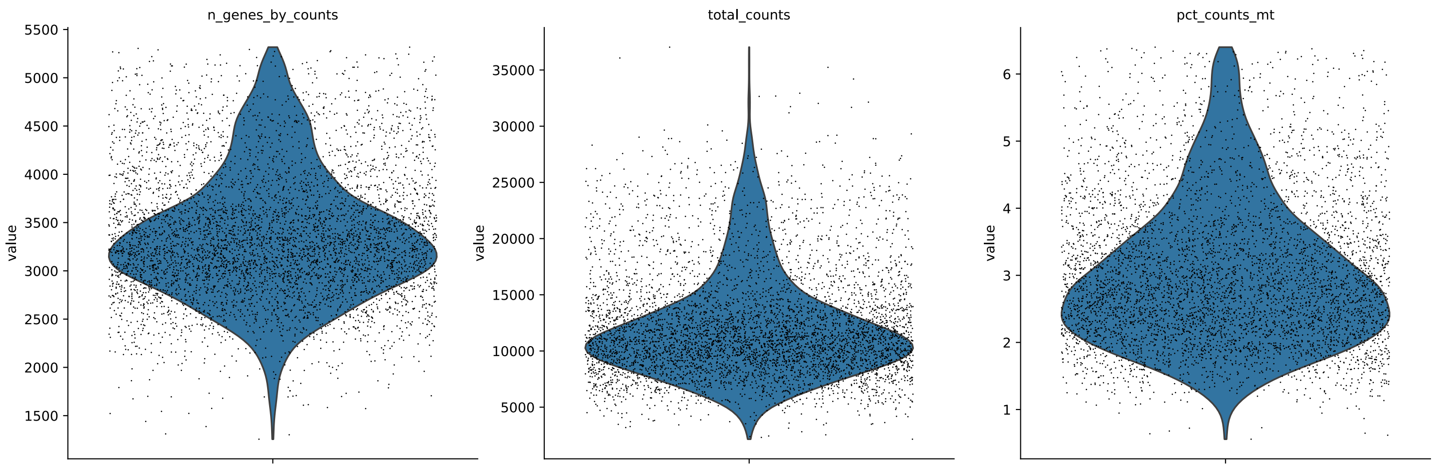
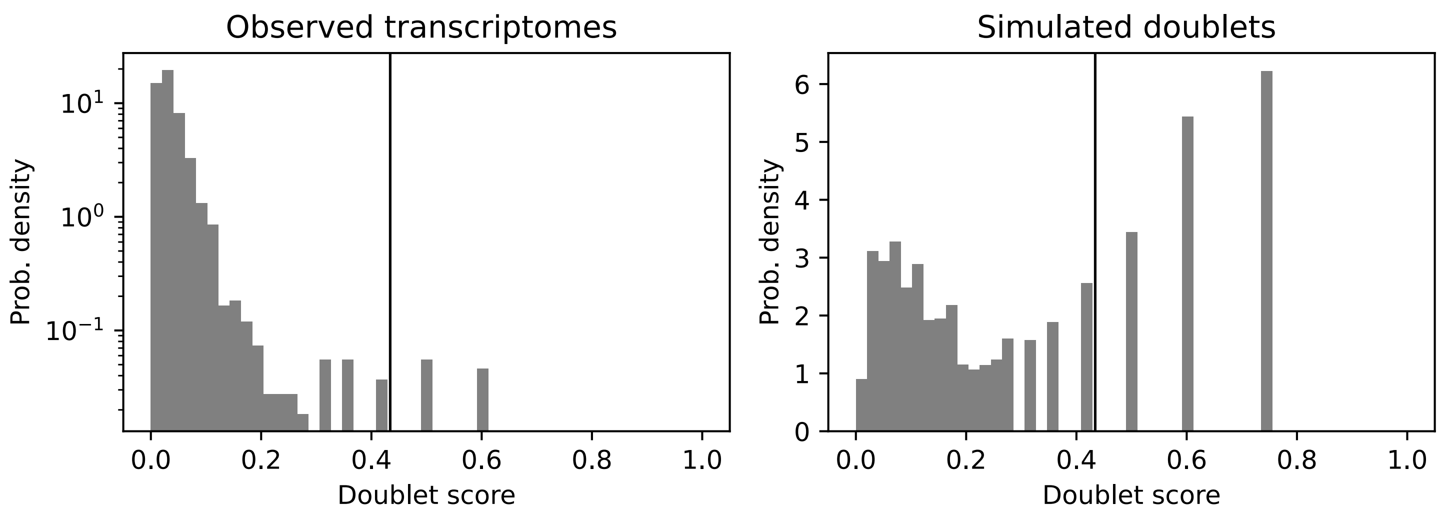
70
80



n_genes_by_counts
total_counts
pct_counts_mt
value
value
value





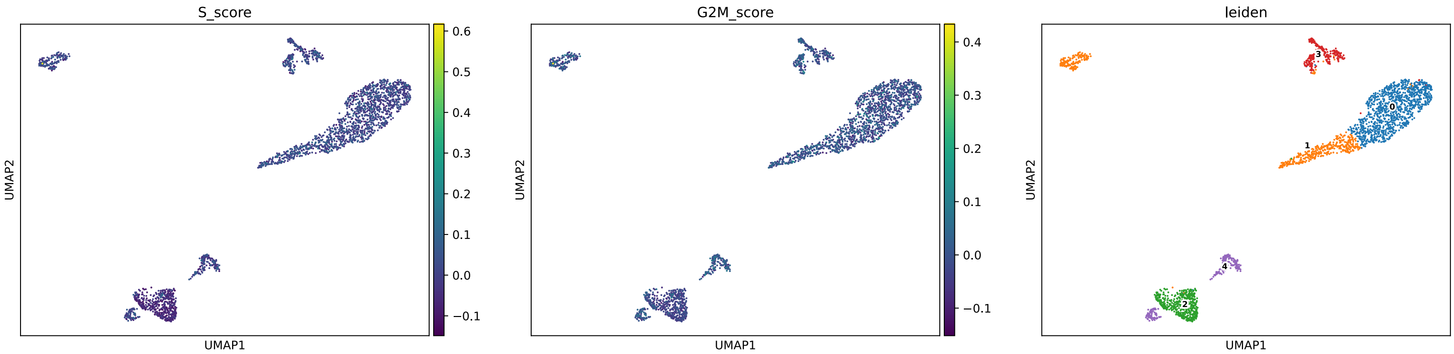


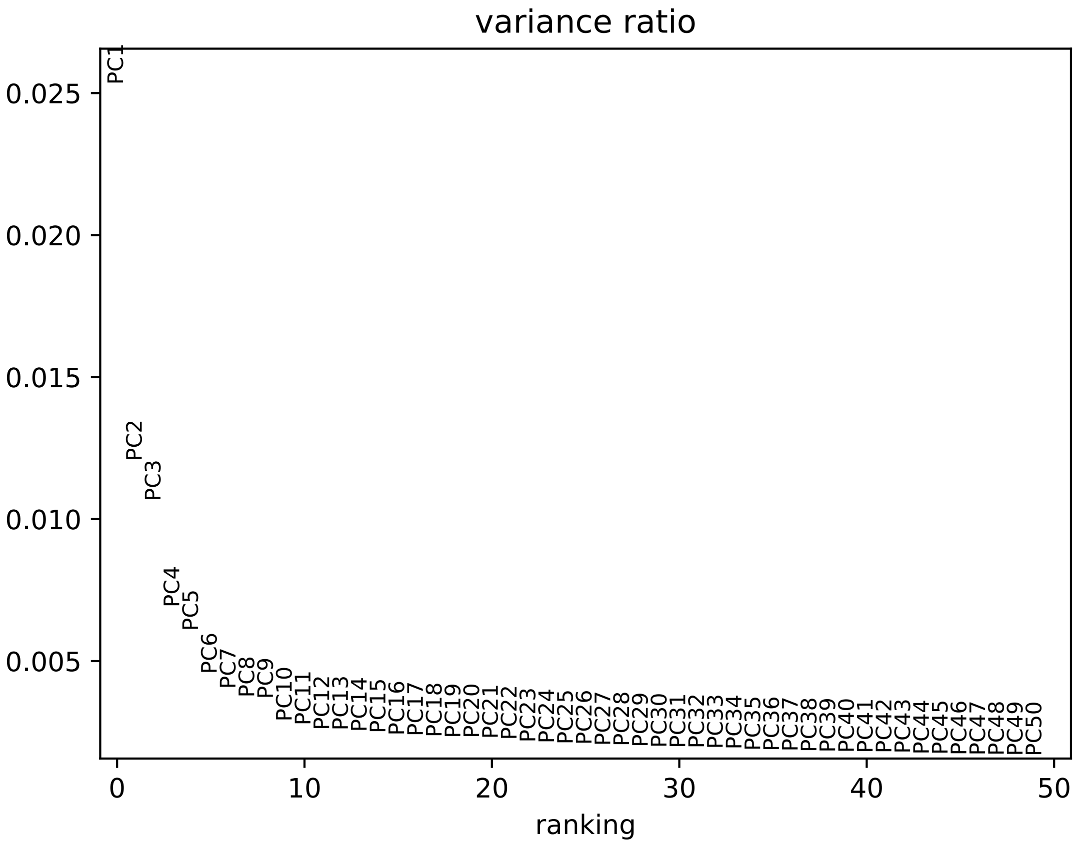
3.2 Dimensionality Reduction and Macro-Clustering

Following initial quality control, the dataset was processed to define the primary biological lineages. The workflow executed a sequential dimensionality reduction followed by graph-based community detection.

Feature Selection and Dimensionality Reduction:

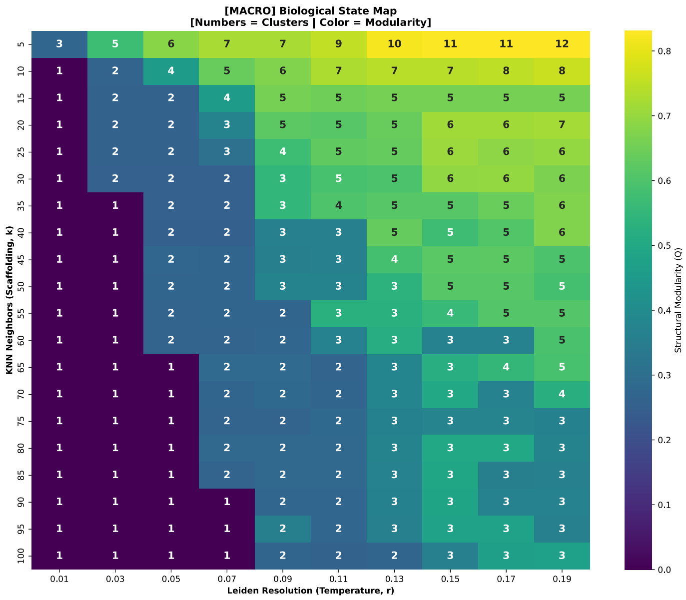
Prior to principal component analysis (PCA), cell cycle scoring (`cell_cycle_check`) was performed to assess whether cell cycle phases were confounding transcriptomic variation. The maximum observed scores (S-phase max = `0.6`, G2M-phase max = `0.4`) indicated minimal phase-driven bias. To ensure clustering was driven by cell identity rather than technical or metabolic variance, `1` mitochondrial and ribosomal genes were explicitly excluded from the Highly Variable Gene (HVG) pool before it was subjected to PCA, which was computed on the remaining `2499` HVGs. An evaluation of the variance explained by each principal component (`evaluation of the PCA variance ratio / elbow plot`) determined that `10` principal components were optimal for capturing the biological signal while minimizing computational noise.

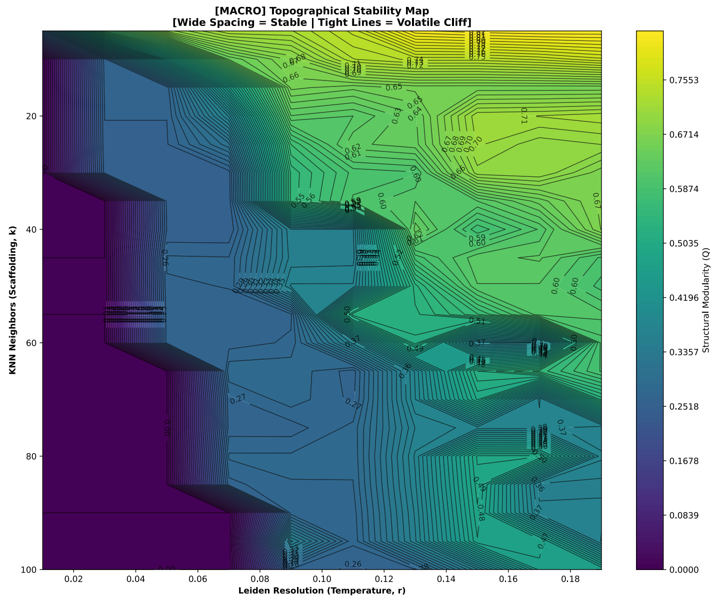


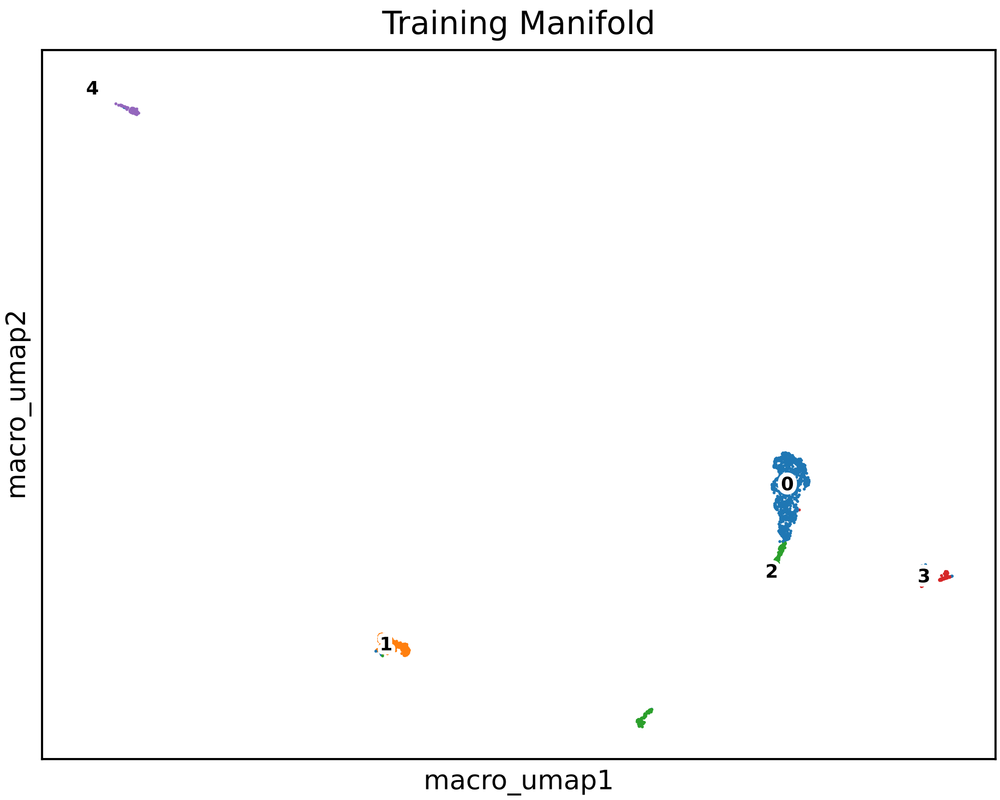


Graph-Based Clustering:

A k-nearest neighbor (kNN) graph and Uniform Manifold Approximation and Projection (UMAP) embeddings were generated (`knn_umap_leiden`). To define robust macro-cluster boundaries objectively, a systematic grid search of the nearest neighbors (k) and Leiden resolution (r) parameters was executed (`macro sweep`). By evaluating clustering stability across iterations (`grid search`) during visual inspection of the parameter space via heat-map and contour plots confirmed this region as highly stable, consistently yielding `5` distinct macro-clusters, the optimal parameters were identified as $k =$ `42` and $r =$ `0.16`. This partitioned the dataset into `5` distinct macro-clusters.





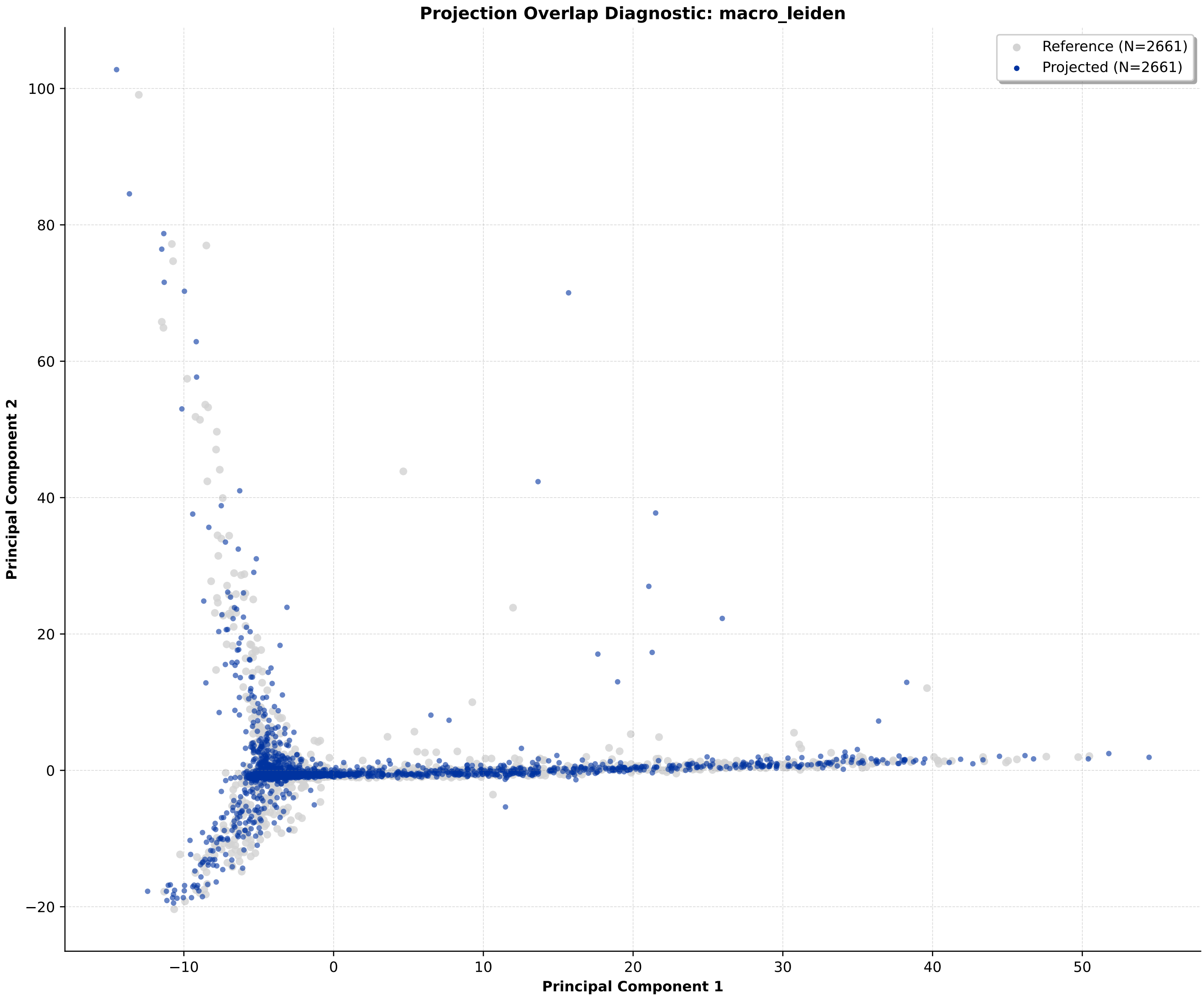


Macro-Cluster Stability Evaluation:

To assess the robustness of the resulting cluster boundaries, a bootstrapping protocol (`jaccard scores`) was applied using 20% data subsampling across repeated iterations. All macro-clusters exceeded the minimum structural stability thresholds. The mean Jaccard survival indices were:

- **Cluster 0:** `0.779` and 1228 cells
- **Cluster 1:** `0.955` and 484 cells
- **Cluster 2:** `0.697` and 399 cells
- **Cluster 3:** `0.890` and 294 cells
- **Cluster 4:** `0.948` and 256 cells

Visual inspection of the resulting projection overlap diagnostic plots confirmed a high degree of structural alignment between the reference ($N = 2,661$) and projected ($N = 2,661$) coordinates. This exact overlap physically demonstrates that the defined micro-cluster boundaries are highly stable and accurately represent the underlying biological topology.

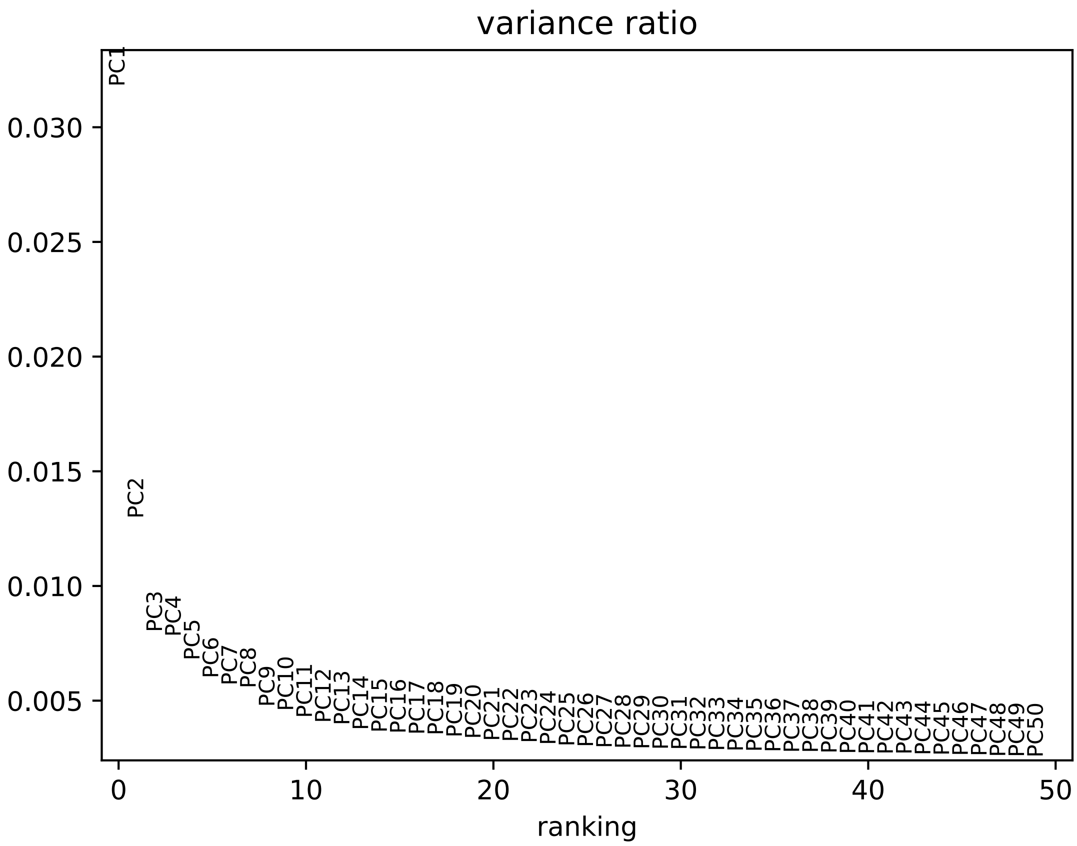


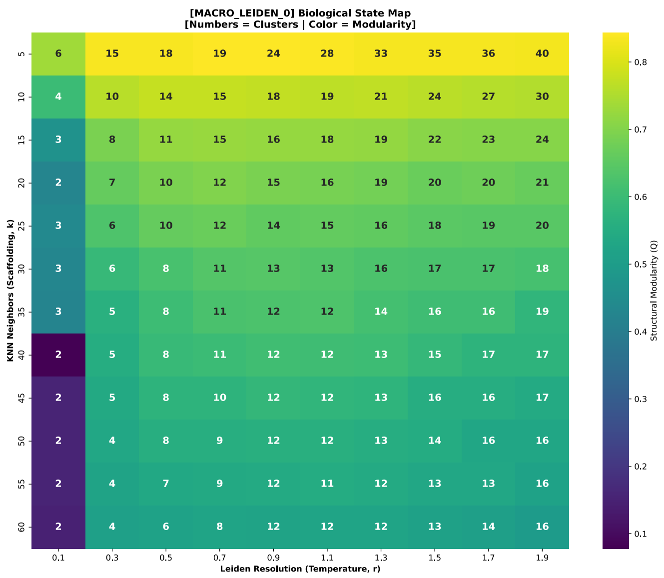
3.3 High-Resolution Micro-Clustering

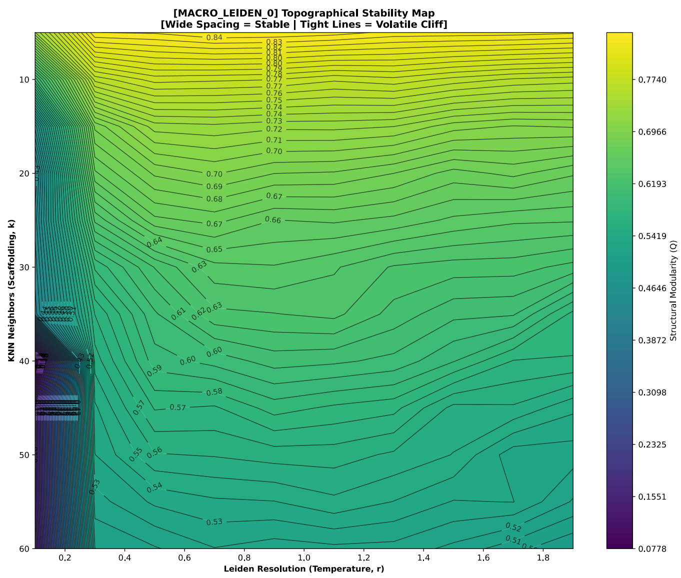
The `5` isolated macro-clusters were independently sub-clustered to identify finer cellular subpopulations (`micro sweep`). Parameter optimization (grid searches for k and r) was performed iteratively for each macro-cluster, yielding a total of `27` distinct sub-clusters across the training dataset.

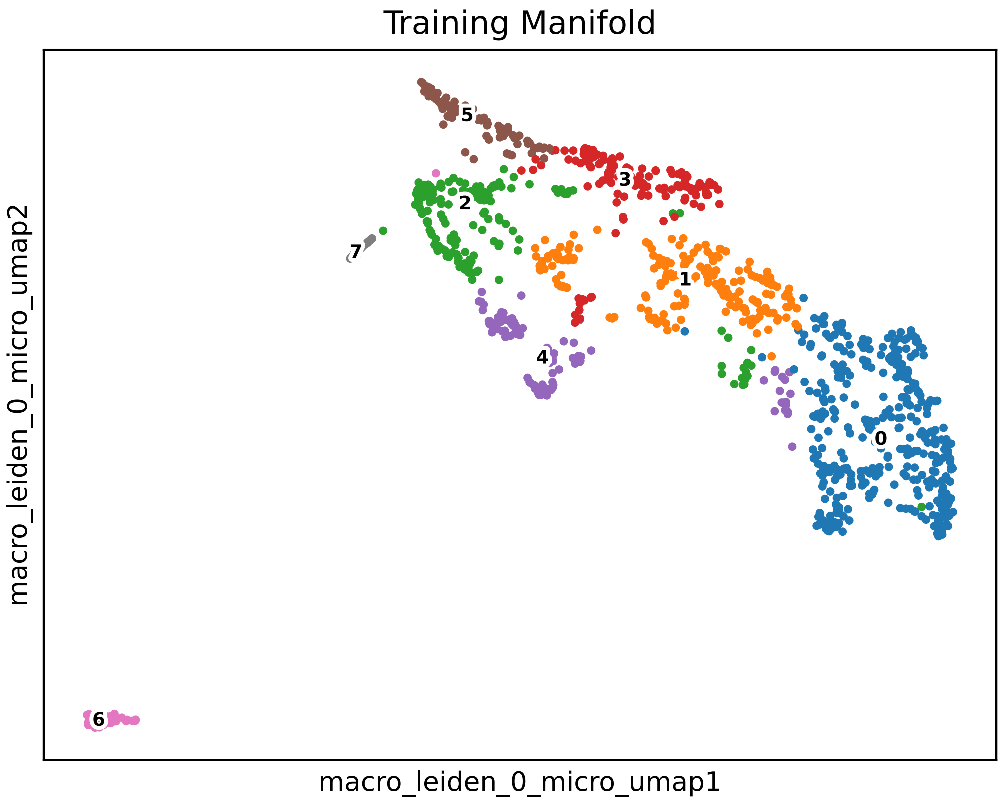
Bootstrapping (`jaccard scores`) was re-applied at the micro-level to evaluate boundary stability. The variance in Jaccard scores successfully differentiated between discrete terminal cell types (high stability, > 0.85) and continuous developmental trajectories (moderate stability, $0.65 - 0.80$):

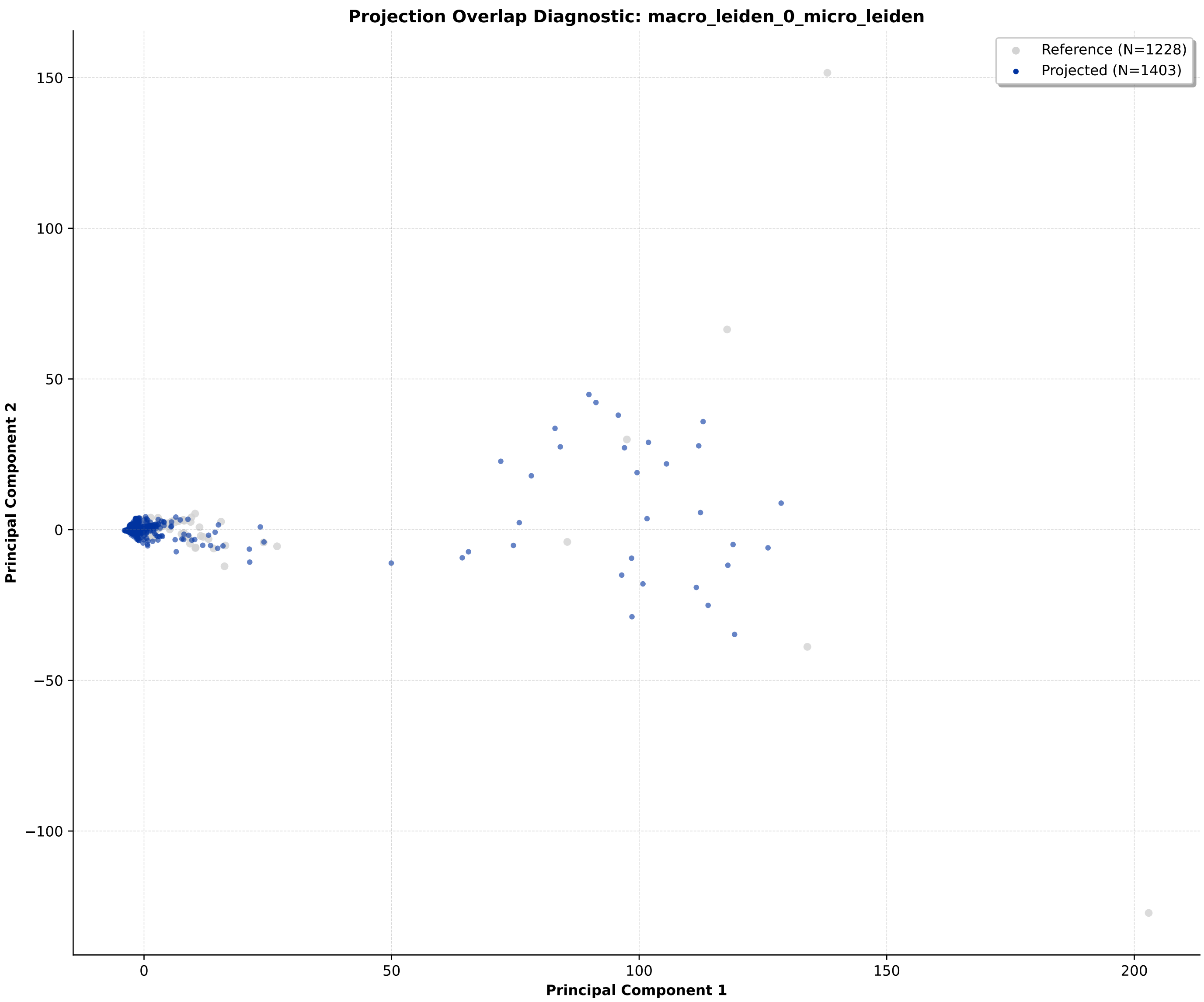
- **Macro-State 0:** Locked at $k =$ `33`, $r =$ `0.46`. Yielded `8` sub-clusters. Minimum observed Jaccard stability: `0.658`.


variance ratio
ranking

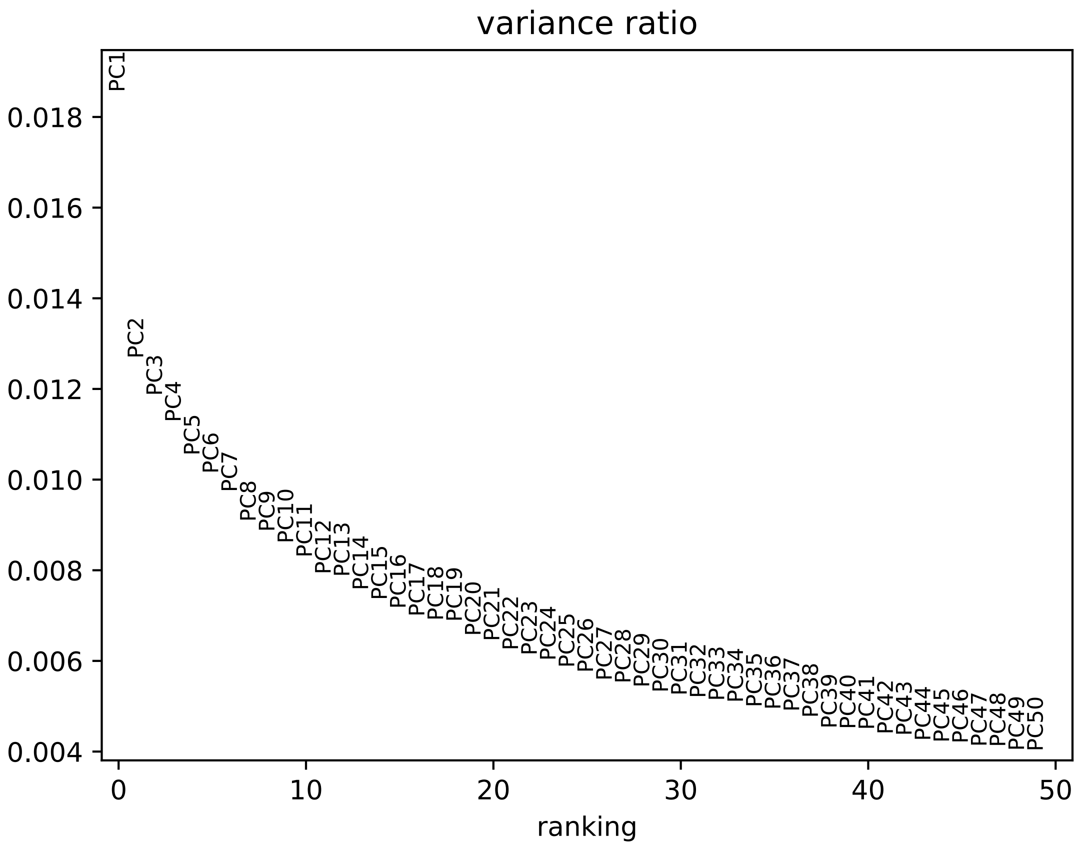

[MACRO_LEIDEN_0] Biological State Map
[Numbers = Clusters | Color = Modularity]
KNN Neighbors (Scaffolding, k)
Leiden Resolution (Temperature, r)
Structural Modularity (Q)

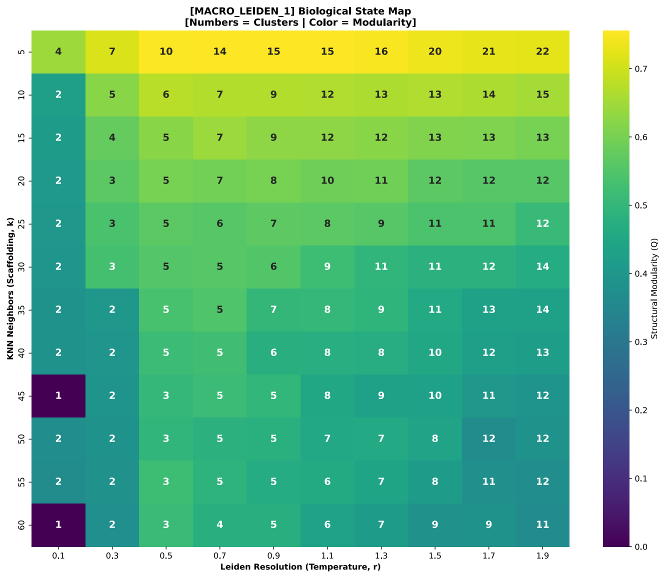

[MACRO_LEIDEN_0] Topographical Stability Map
[Wide Spacing = Stable | Tight Lines = Volatile Cliff]
KNN Neighbors (Scaffolding, k)
Leiden Resolution (Temperature, r)
Structural Modularity (Q)

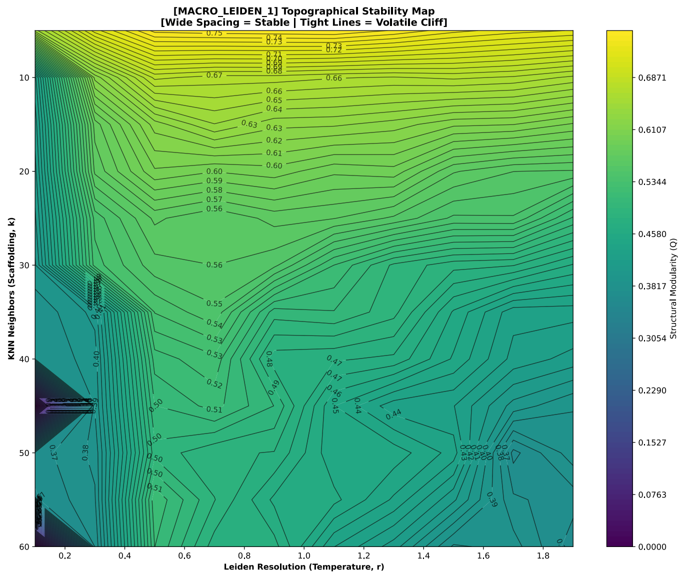


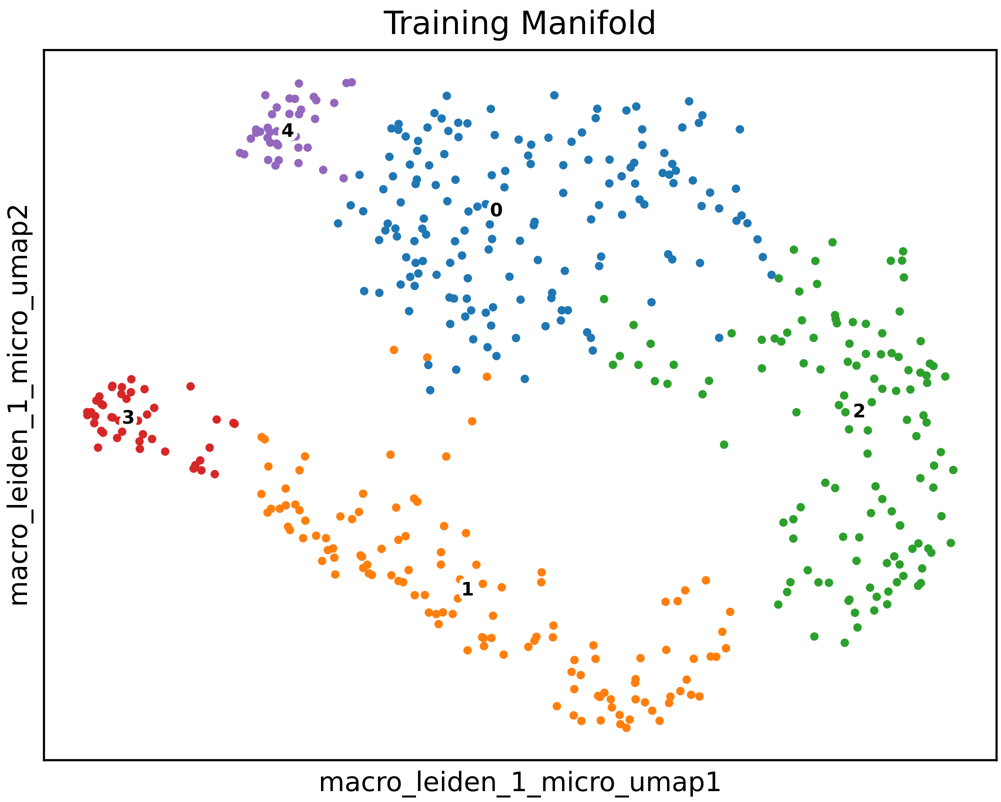


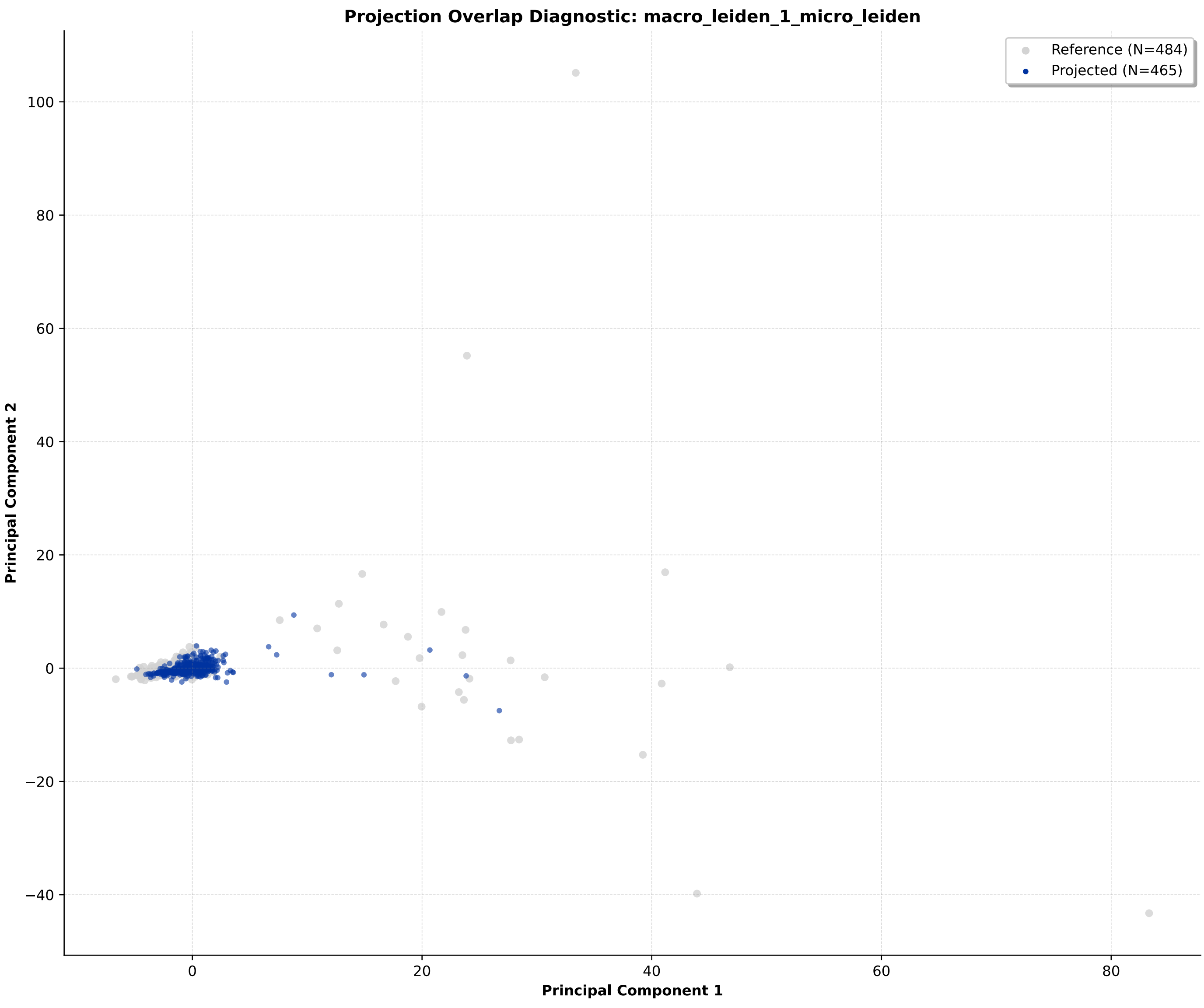
- **Macro-State 1:** Locked at $k = $ `32`, $r = $ `0.68`. Yielded `5` sub-clusters. Minimum observed Jaccard stability: `0.797`.


variance ratio
ranking

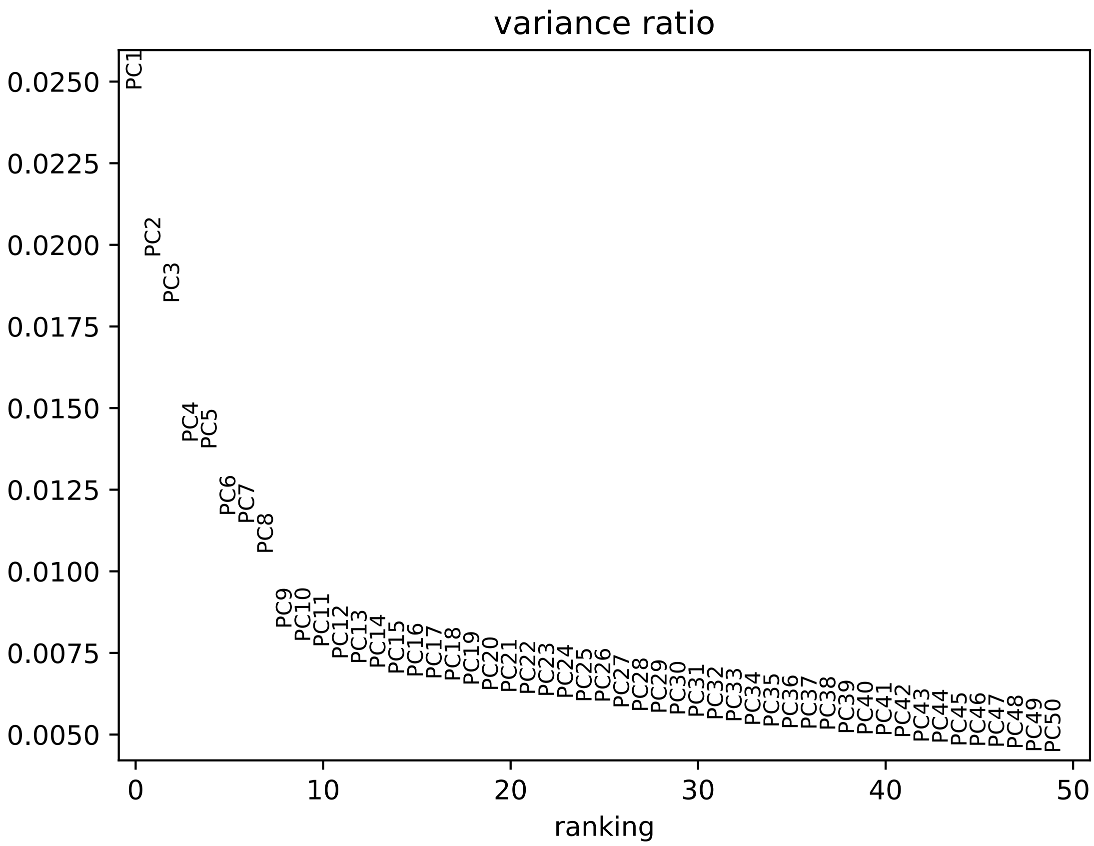

[MACRO_LEIDEN_1] Biological State Map
[Numbers = Clusters | Color = Modularity]
KNN Neighbors (Scaffolding, k)
Leiden Resolution (Temperature, r)
Structural Modularity (Q)

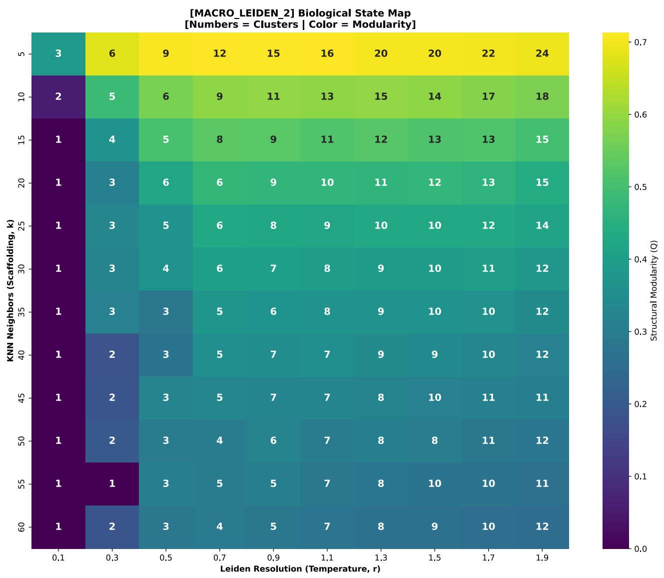

[MACRO_LEIDEN_1] Topographical Stability Map
[Wide Spacing = Stable | Tight Lines = Volatile Cliff]
KNN Neighbors (Scaffolding, k)
Leiden Resolution (Temperature, r)
Structural Modularity (Q)

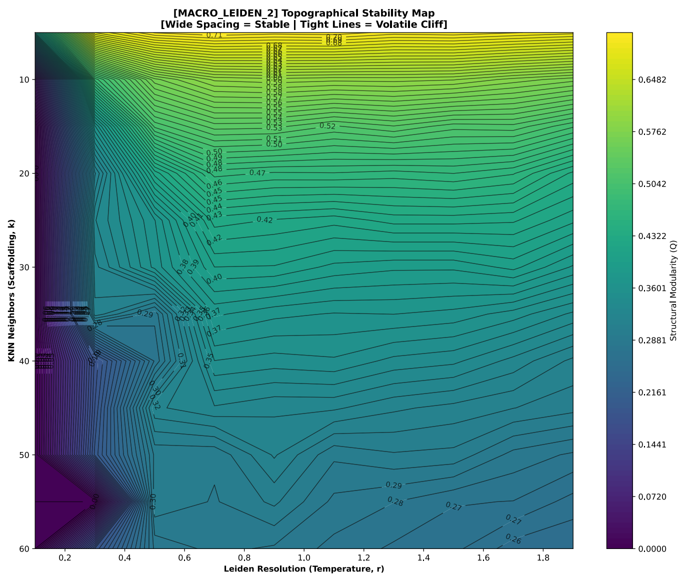


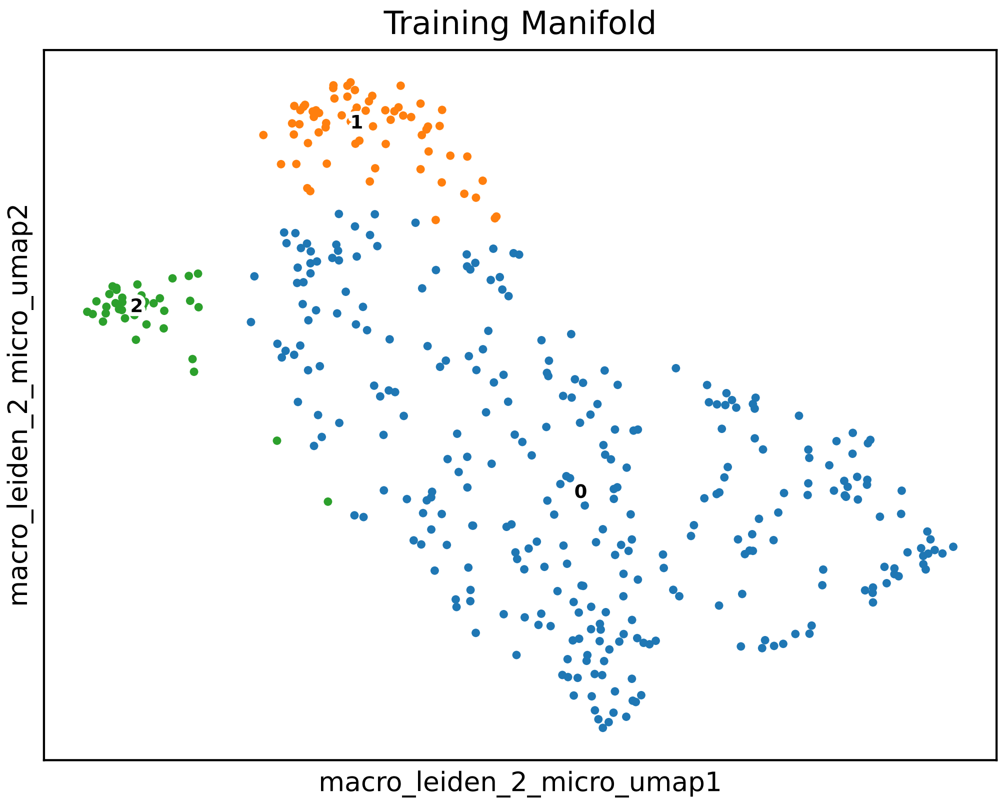


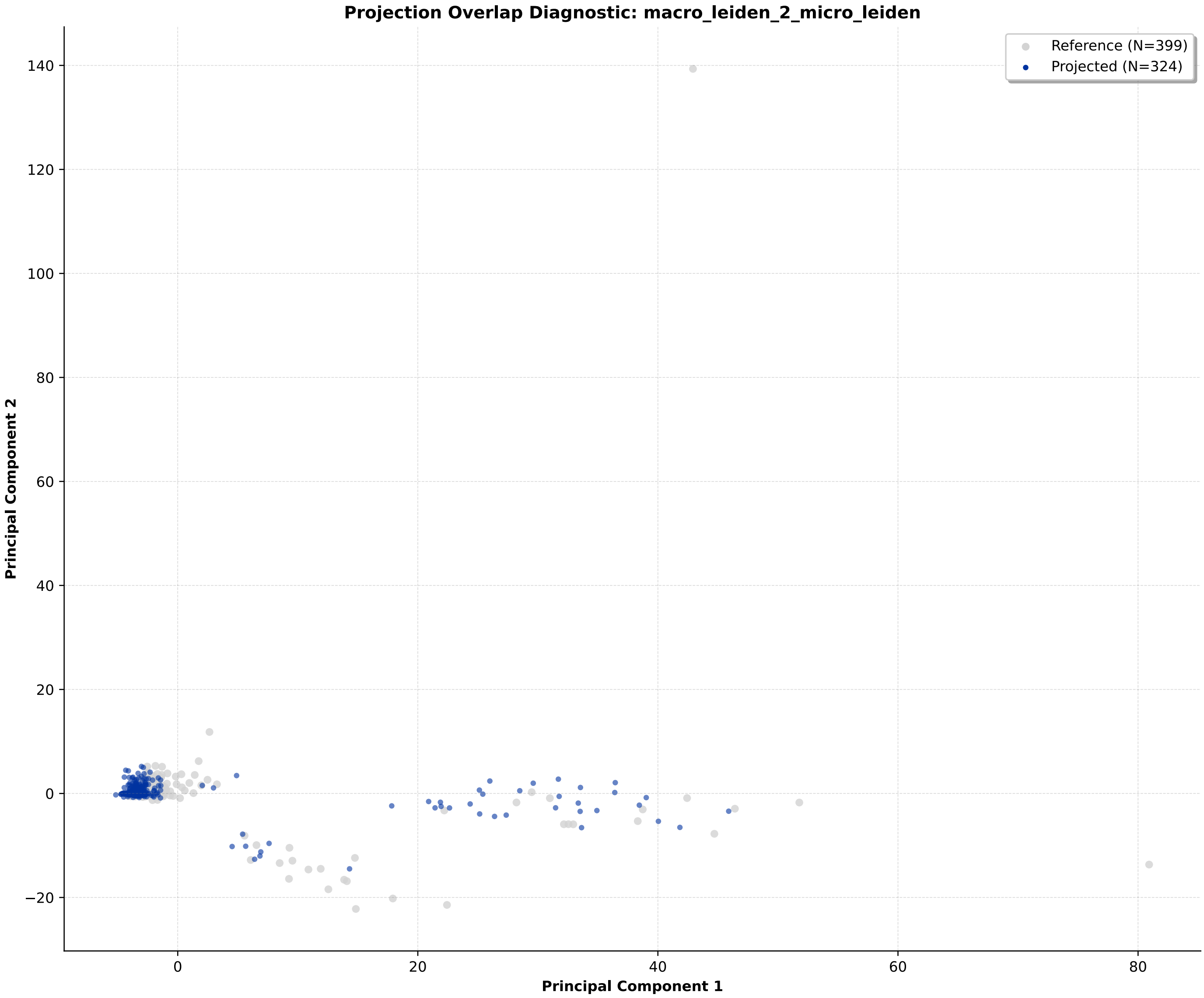
- **Macro-State 2:** Locked at $k = 35$, $r = 0.44$. Yielded `3` sub-clusters. Minimum observed Jaccard stability: `0.743`.


variance ratio
ranking

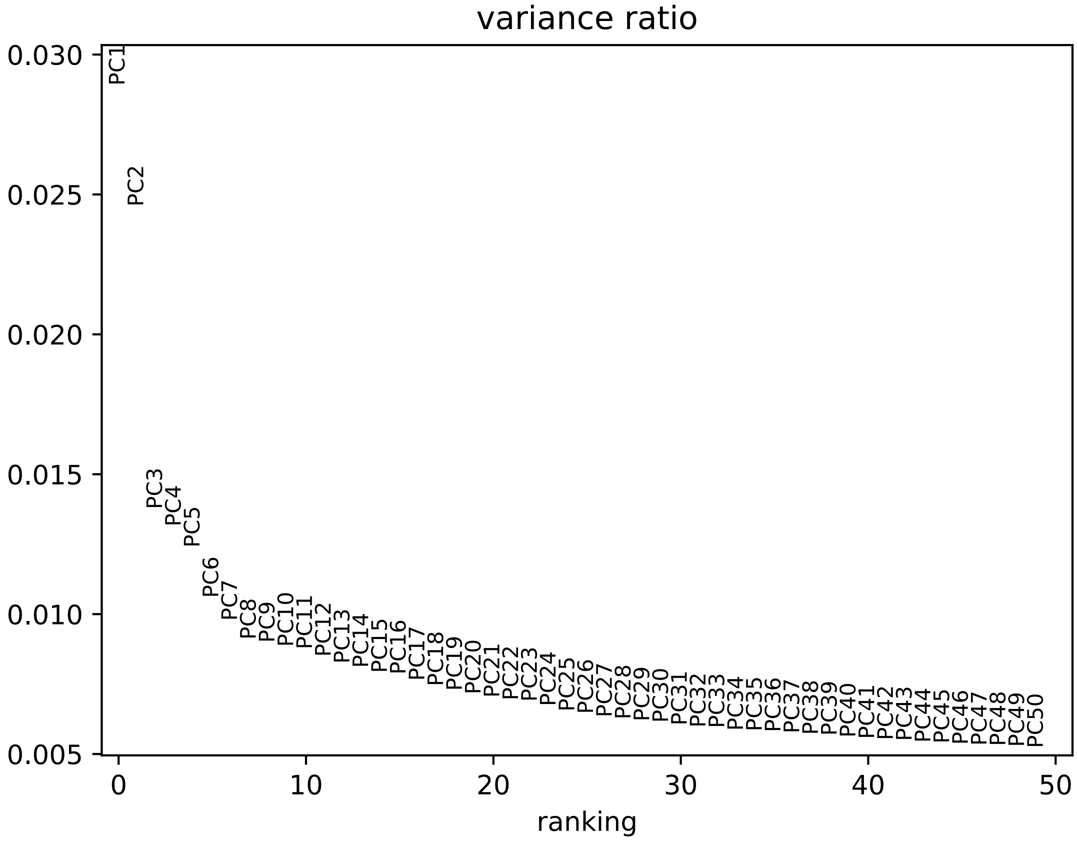

[MACRO_LEIDEN_2] Biological State Map
[Numbers = Clusters | Color = Modularity]
KNN Neighbors (Scaffolding, k)
Leiden Resolution (Temperature, r)
Structural Modularity (Q)


[MACRO_LEIDEN_2] Topographical Stability Map
[Wide Spacing = Stable | Tight Lines = Volatile Cliff]
KNN Neighbors (Scaffolding, k)
Leiden Resolution (Temperature, r)
Structural Modularity (Q)

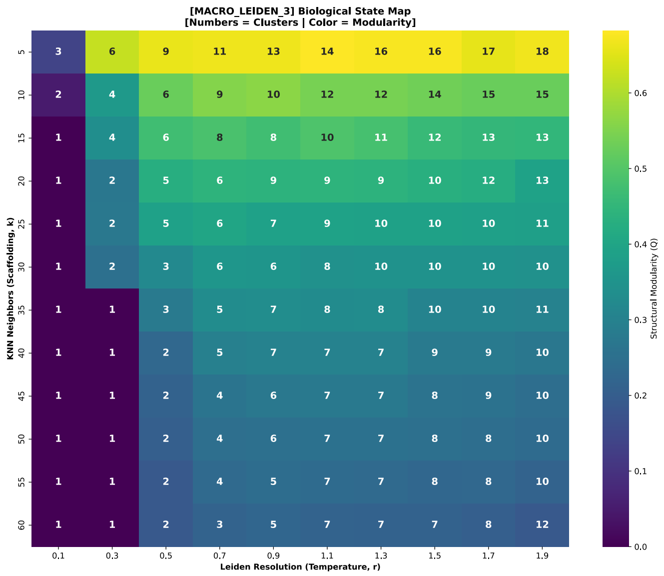




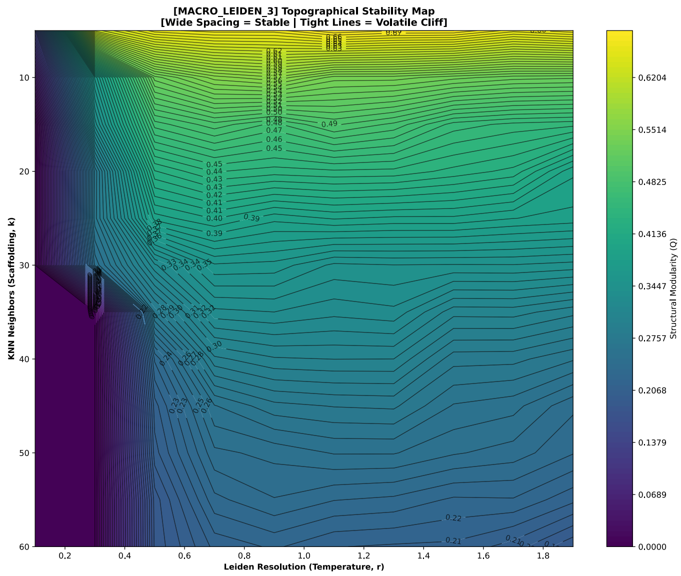
- **Macro-State 3:** Locked at $k = 50$, $r = 1.2$. Yielded `7` sub-clusters. Minimum observed Jaccard stability: `0.660`.



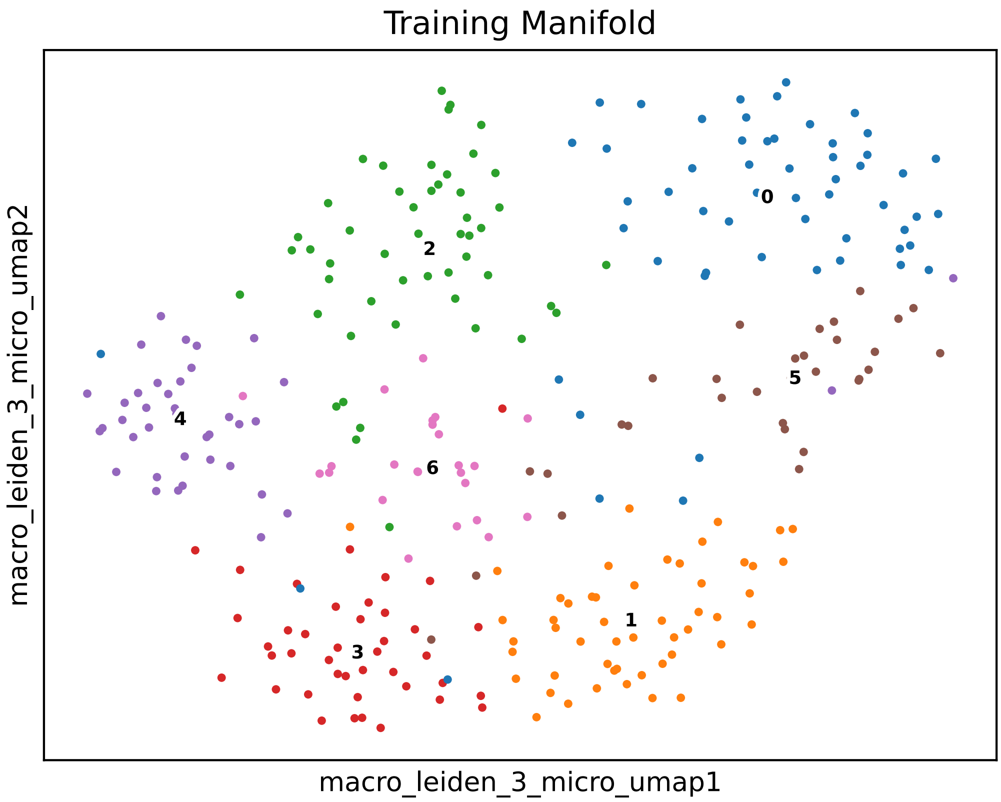
variance ratio
ranking

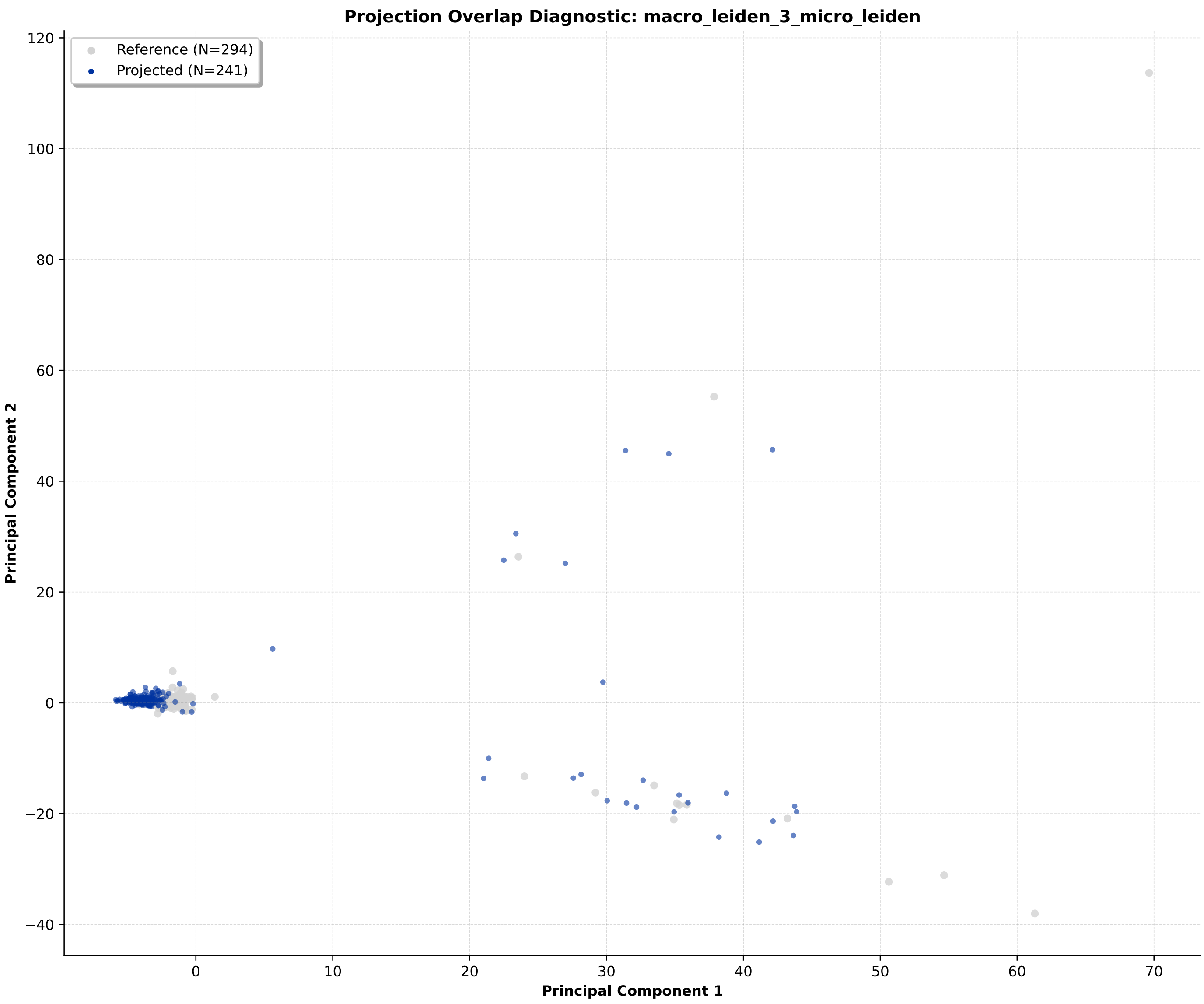


[MACRO_LEIDEN_3] Biological State Map
[Numbers = Clusters | Color = Modularity]
KNN Neighbors (Scaffolding, k)
Leiden Resolution (Temperature, r)
Structural Modularity (Q)

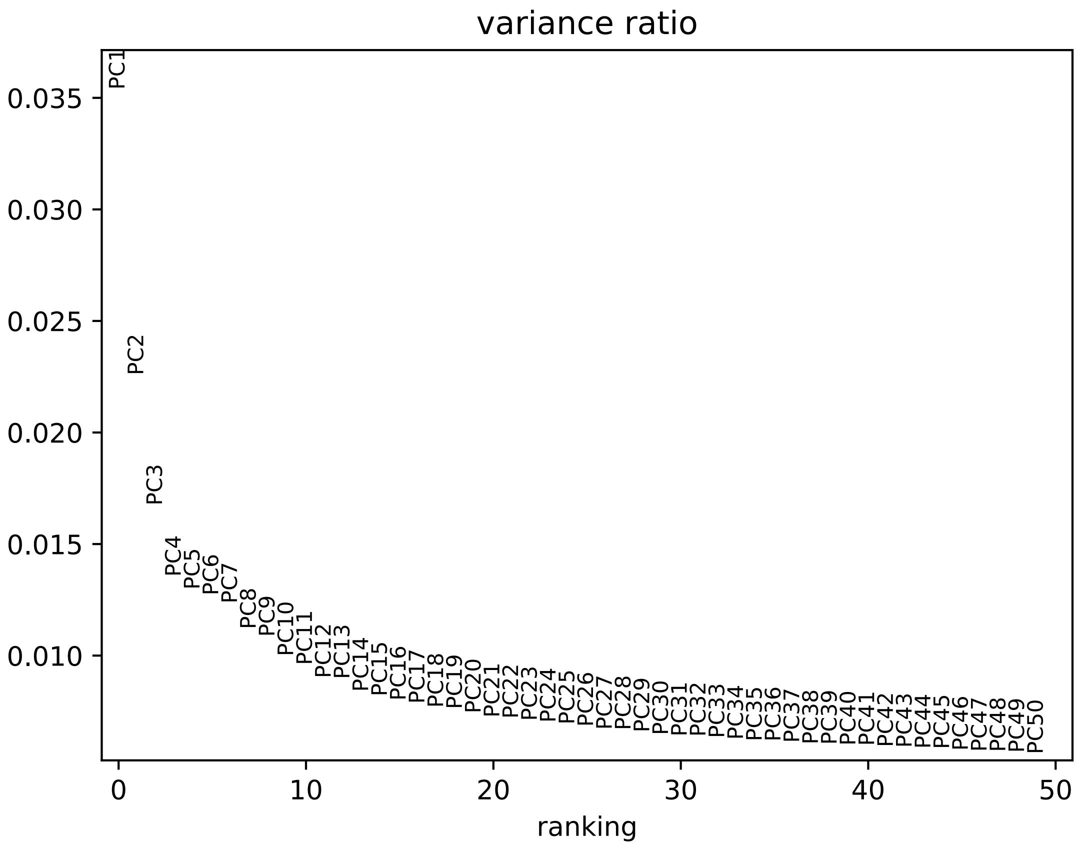


[MACRO_LEIDEN_3] Topographical Stability Map
[Wide Spacing = Stable | Tight Lines = Volatile Cliff]
KNN Neighbors (Scaffolding, k)
Leiden Resolution (Temperature, r)
Structural Modularity (Q)





- **Macro-State 4:** Locked at $k = $ `25`, $r = $ `0.5`. Yielded `4` sub-clusters. Minimum observed Jaccard stability: `0.756`.


variance ratio
0.035
0.030
0.025
0.020
0.015
0.010
0
10
20
30
40
50
ranking
PC1
PC2
PC3
PC4
PC5
PC6
PC7
PC8
PC9
PC10
PC11
PC12
PC13
PC14
PC15
PC16
PC17
PC18
PC19
PC20
PC21
PC22
PC23
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PC25
PC26
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PC30
PC31
PC32
PC33
PC34
PC35
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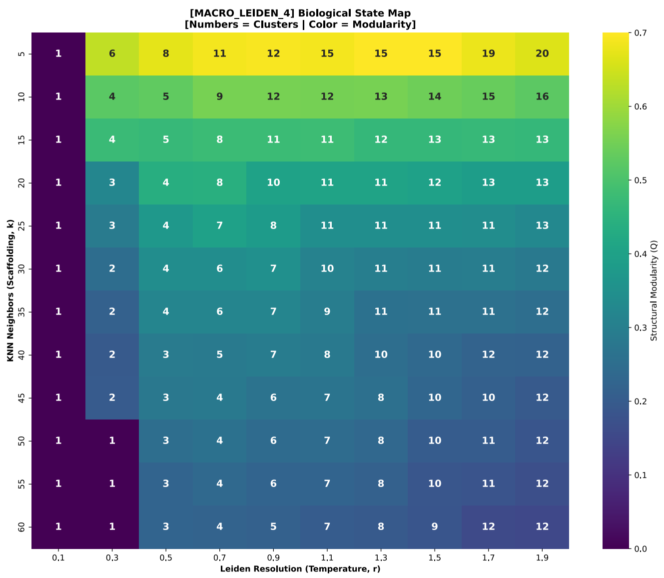
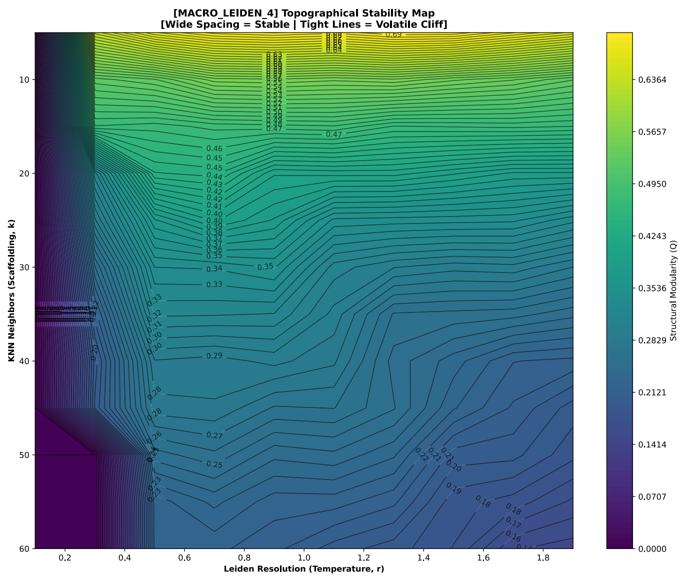
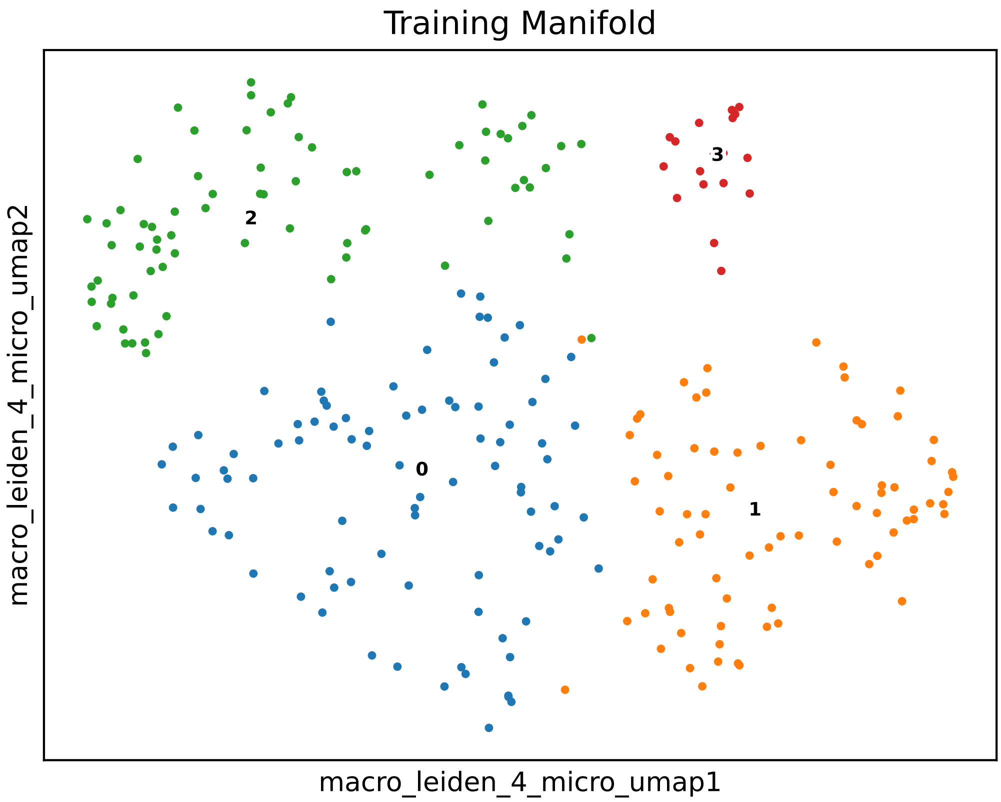
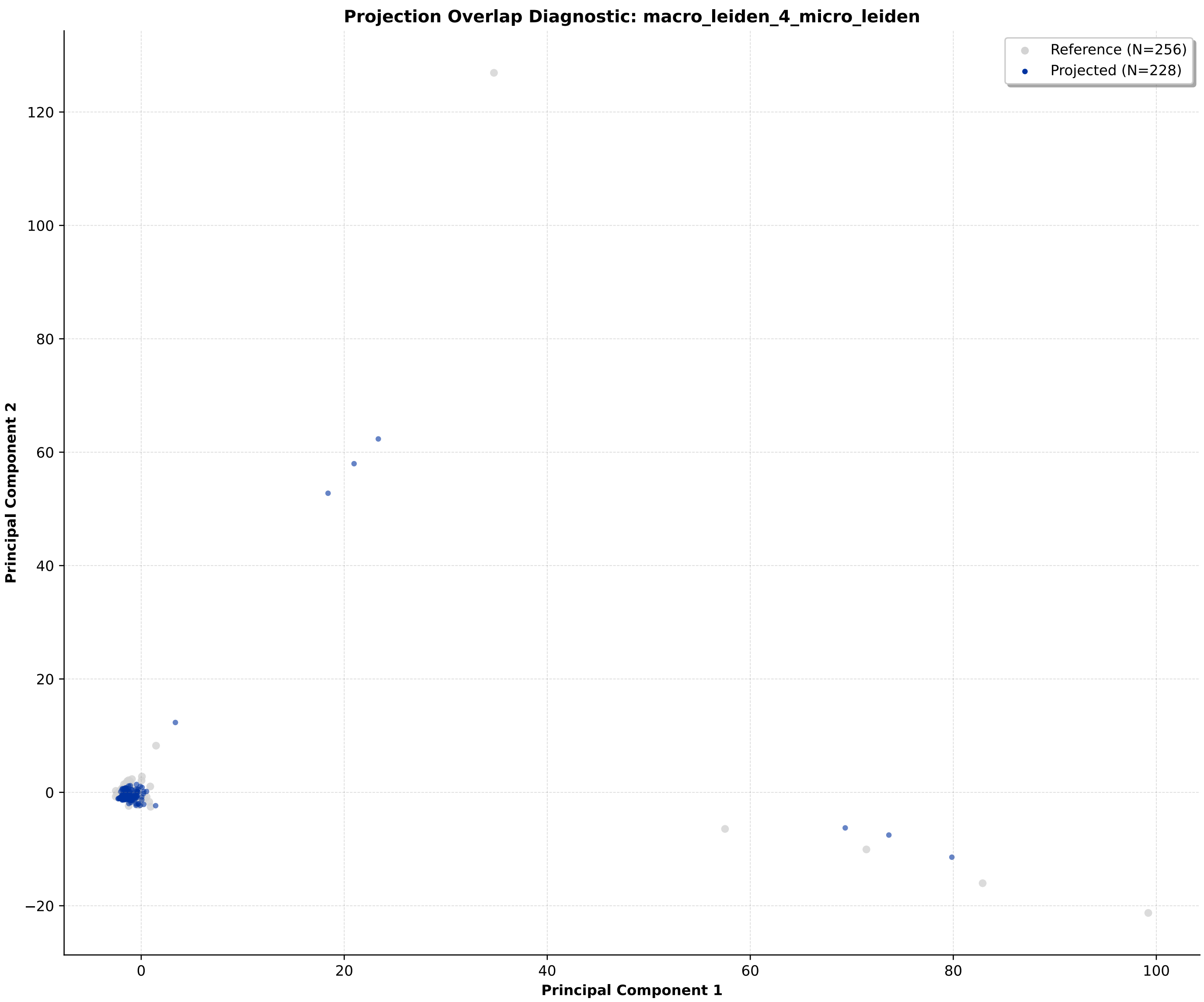
PC45
PC46
PC47
PC48
PC49
PC50


[MACRO_LEIDEN_4] Biological State Map
[Numbers = Clusters | Color = Modularity]
KNN Neighbors (Scaffolding, k)
Leiden Resolution (Temperature, r)
0,1
0,3
0,5
0,7
0,9
1,1
1,3
1,5
1,7
1,9
5: 1 6 8 11 12 15 15 15 19 20
10: 1 4 5 9 12 12 13 14 15 16
15: 1 4 5 8 11 11 12 13 13 13
20: 1 3 4 8 10 11 11 12 13 13
25: 1 3 4 7 8 11 11 11 11 13
30: 1 2 4 6 7 10 11 11 11 12
35: 1 2 4 6 7 9 11 11 11 12
40: 1 2 3 5 7 8 10 10 12 12
45: 1 2 3 4 6 7 8 10 10 12
50: 1 1 3 4 6 7 8 10 11 12
55: 1 1 3 4 6 7 8 10 11 12
60: 1 1 3 4 5 7 8 9 12 12
Structural Modularity (Q)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7


[MACRO_LEIDEN_4] Topographical Stability Map
[Wide Spacing = Stable | Tight Lines = Volatile Cliff]
KNN Neighbors (Scaffolding, k)
Leiden Resolution (Temperature, r)
0,2
0,4
0,6
0,8
1,0
1,2
1,4
1,6
1,8
Structural Modularity (Q)
0.0000
0.0707
0.1414
0.2121
0.2829
0.3536
0.4243
0.4950
0.5657
0.6364





3.4 Differential Gene Expression and Marker Extraction

Following clustering, differential gene expression analysis was performed to identify cluster-specific marker genes for cell type annotation.

Differential Expression Analysis: differential gene expression (DGE) profiling (`rank_genes_group`) was executed across the validated topology using the Wilcoxon rank-sum test. For each defined cluster, candidate marker genes were evaluated using a dual-metric approach: statistical significance via FDR-adjusted p-values (`pvals_adj`) and spatial expression exclusivity (`expression_delta`). To be considered as a candidate marker, a gene was required to meet a baseline significance threshold of an adjusted p-value < `0.05` and a minimum $\log_2$ fold-change of `1.0` . The most discriminative features (`top_genes`) were extracted to define the transcriptomic signature of each cluster.

Cluster Filtering for Statistical Robustness: To prevent false-positive annotations resulting from low cell counts or extreme data sparsity, an adaptive filtering threshold (`elastic threshold`) was applied to the DGE outputs. Clusters failing to meet the minimum threshold for cellular representation ($N \geq$ `10` cells) or lacking sufficient statistical confidence in their marker profiles were flagged as non-viable. This filtering retained total of `24` high-confidence (`viable_states`) sub-clusters for downstream annotation.

Marker Extraction Metrics:

- **Major Cell Lineages:** Distinct macro-clusters exhibited strong transcriptional divergence. Marker genes defining these major lineages recorded adjusted p-values approaching the computational minimum (`1.0e324`) and high spatial exclusivity, with `expression_delta` scores exceeding :
  - **Cluster 0:** `0.544`
  - **Cluster 1:** `0.915`
  - **Cluster 2:** `0.135`
  - **Cluster 3:** `0.560`
  - **Cluster 4:** `0.681`
- **Sub-cluster Resolution:** Within closely related sub-clusters, highly specific markers demonstrated reduced `expression_delta` scores due to shared baseline expression. Marker genes distinguishing these related sub-types recorded `expression_delta` values ranging from:
  - **Cluster 0:** `0.144` to `0.799`
  - **Cluster 1:** `0.235` to `0.877`
  - **Cluster 2:** `0.388` to `0.807`
  - **Cluster 3:** `0.277` to `0.954`
  - **Cluster 4:** `0.225` to `0.527`

Independent Marker Validation: Because standard DGE statistics rely on relative contrast, which can obscure shared functional markers in continuous sub-lineages, candidate signatures were subjected to an independent validation protocol (`absence_cross_validation`). The absolute expression profiles of canonical lineage markers, curated from `Teichmann Lab. (2023). _Basic_celltype_information.xlsx_ Data file. CellTypist Wiki: Pan−Immune CellTypist Atlas v2. Available at: https://github.com/Teichlab/celltypist_wiki/blob/main/atlases/Pan_Immune_CellTypist/v2/tables/Basic_celltype_information.xlsx.` , were quantitatively and visually evaluated across all clusters (`wide_span_plots`). This orthogonal cross-validation confirmed the true biological presence or absence of specific genes, successfully resolving ambiguous cell states where continuous background expression locally suppressed primary DGE significance.

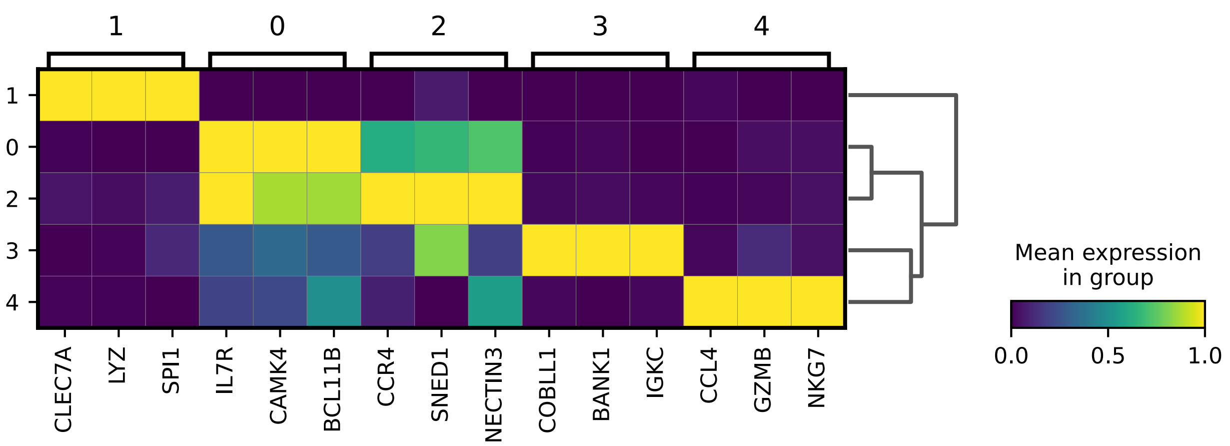
3.5 Phenotype Annotation and Tensor Recombination

Following marker gene extraction, biological identities were assigned to the identified sub-clusters. These annotations were then mapped to standardized ontologies, and the partitioned datasets were recombined into a single, fully annotated object.

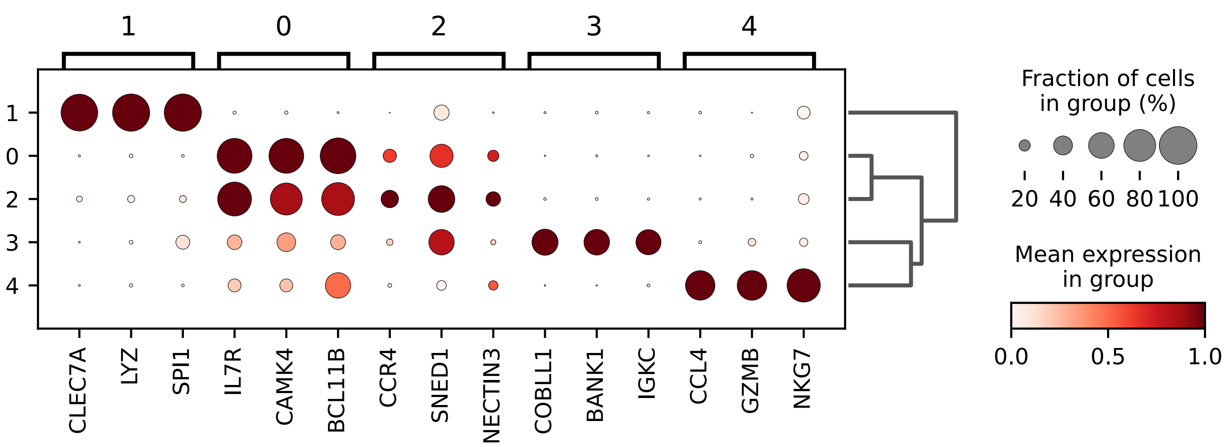
3.5.1 Phenotype Annotation and Cell Ontology Mapping

Human-derived biological annotations, guided by the statistical marker profiles (Section 3.4), were systematically injected into the dataset metadata (`orc_annotation`). To ensure interoperability and standardization, all micro-state annotations were mapped to the formal Cell Ontology (CL) reference structure.

matrixplot_macro_leiden_top_genes



dotplot_macro_leiden_top_genes

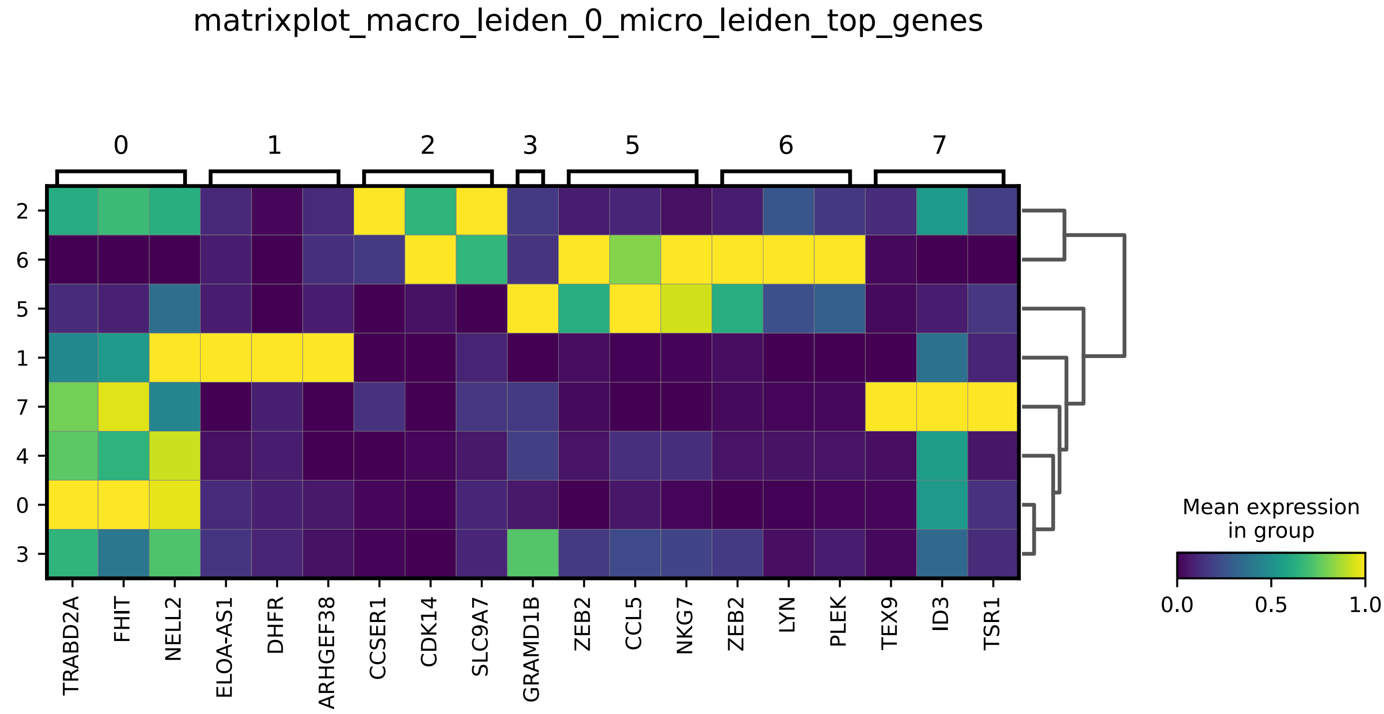


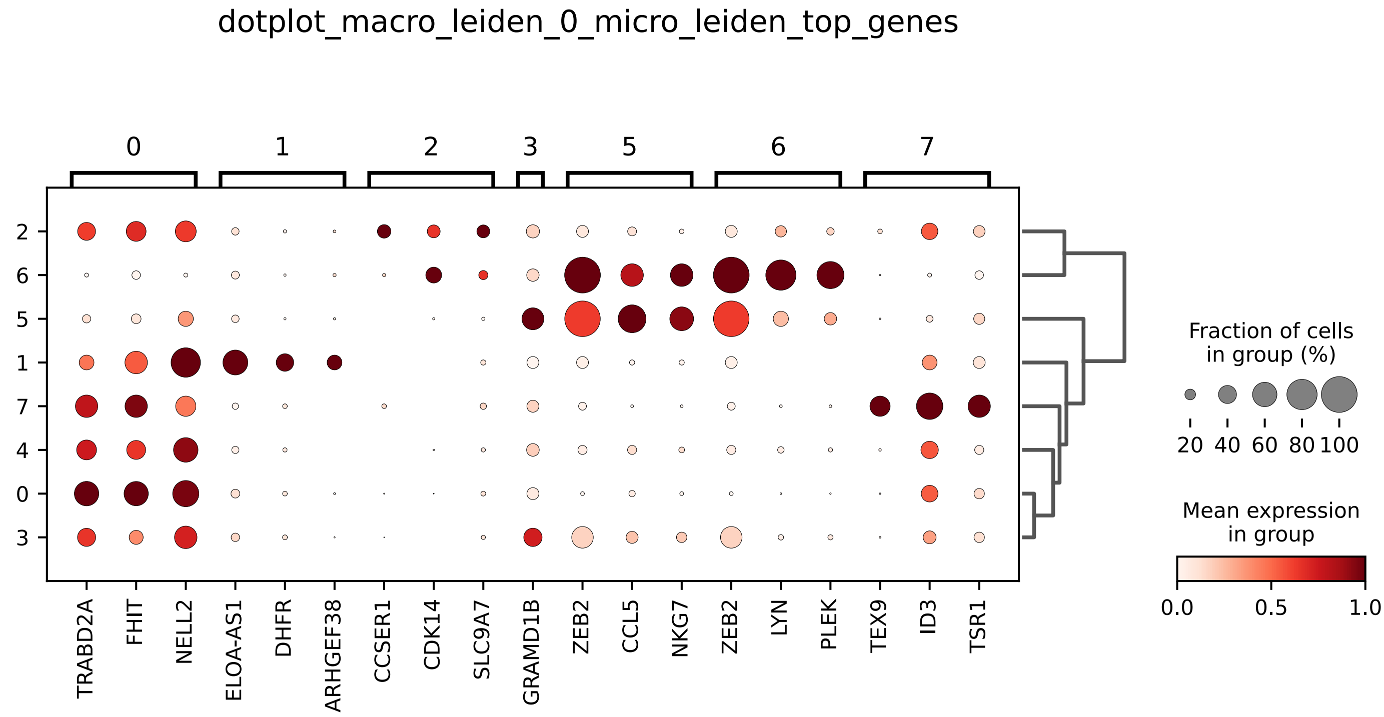
The hierarchical annotation of the dataset is detailed below:

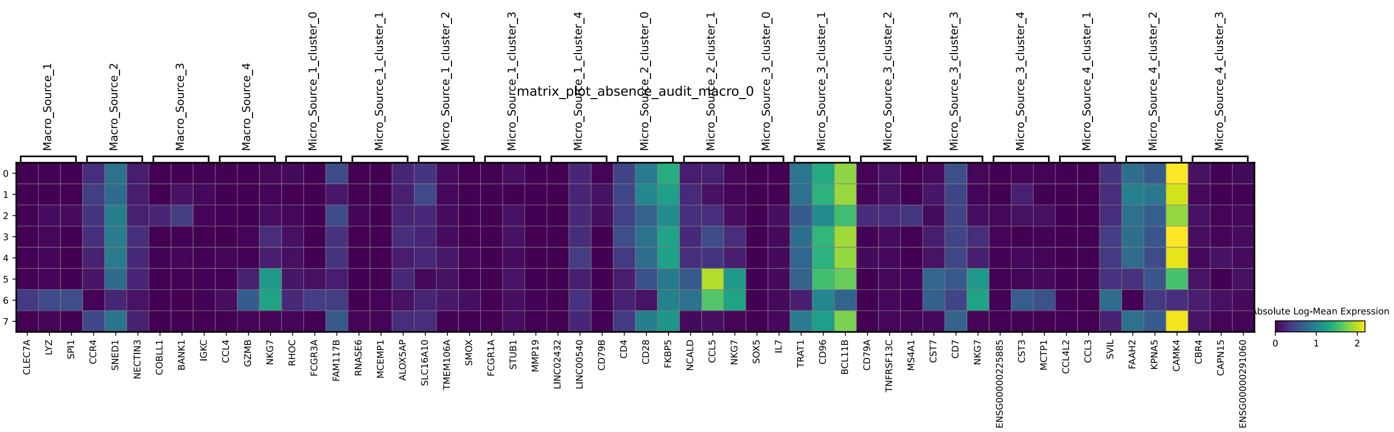
Macro-Cluster 0: `T-Cell Lineage (CL:0000084)`

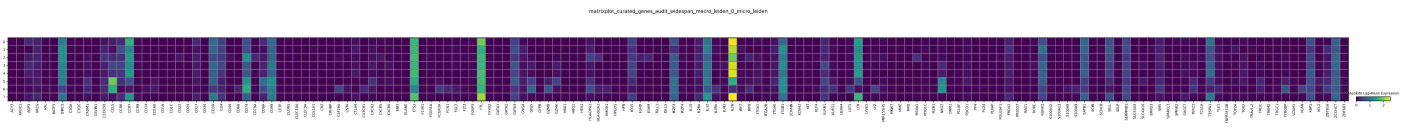
- **Lineage Markers:** `High expression of the canonical T-cell transcription factor BCL11B, coupled with high baseline expression of IL7R and CAMK4. The cluster was topologically partitioned from Macro-Cluster 2 despite shared baseline transcription, evidenced by a moderate (0.4 mean) expression of CCR4, SNED1, and`

`NECTIN3..`









- **Micro-Cluster 0.0:** `Naive T-Cell (CL:0000898)`
  - **Lineage Markers:** `Retention of the foundational T-cell manifold signature (high CAMK4, moderate BCL11B) establishes structural continuity with the`

parent macro-cluster. The clustering boundary of this sub-state is driven by the exclusive, high-level co-expression of TRABD2A and FHIT, tightly coupled with NELL2—a canonical marker of resting lymphocytes. Supported by high absolute expression of the homeostatic survival receptor IL7R, this signature isolates the resting, Naive T-Cell compartment..

- **Micro-Cluster 0.1:** `Early Activated / Proliferating T-Cell (CL:0000899)`
  - **Lineage Markers:** `Anchored to the parent T-cell manifold via high CAMK4 and moderate BCL11B, this sub-state maintains strict topological adjacency to the naive pool (Micro-Cluster 0.0) through sustained homeostatic IL7R expression and residual NELL2 transcription. However, the clustering boundary isolating this cluster is driven entirely by a functional metabolic shift. The high, exclusive upregulation of DHFR—an enzyme physically required for de novo nucleotide biosynthesis and clonal expansion—coupled with the cytoskeletal remodeling factor ARHGEF38 and the transcript ELOA-AS1, isolates a state of early cellular activation and active proliferation diverging from the resting naive baseline.`
- **Micro-Cluster 0.2:** `Transitional / Memory T-Cell (CL:0000813)`
  - **Lineage Markers:** `Topological divergence from the naive state is evidenced by the transcriptomic attenuation of the foundational homeostatic anchors, with IL7R and CAMK4 dropping to moderate and low-moderate expression tiers, respectively. The discrete boundaries of this sub-state are established by a signature of cellular priming and organelle remodeling. The cluster is uniquely defined by the positive expression of SLC9A7—indicating targeted pH regulation of the Golgi apparatus for altered secretory trafficking—alongside CCSER1 and a low-amplitude, high-variance expression of CDK14. This profile isolates a structurally poised, non-proliferative transitional or memory T-cell compartment that has exited strict naive homeostasis but maintains developmental continuity with the BCL11B-positive parent manifold.`
  - **Micro-Cluster 0.3:** `Metabolically Poised Transitional T-Cell (CL:0000813)`
    - **Lineage Markers:** `Sustained high expression of IL7R dictates an ongoing reliance on peripheral homeostatic survival networks. However, The downregulation of CAMK4 and BCL11B separates this population from the naive state. This sub-cluster is characterized by high-variance expression of GRAMD1B. Despite a low mean expression tier (0.5), this sterol-transport transcript indicates targeted biophysical remodeling of plasma membrane lipid rafts. This signature identifies a distinct, metabolically poised transitional T-cell pool undergoing structural membrane reorganization, indicative of memory lineage commitment or immune synapse priming.`
- **Micro-Cluster 0.4:** `Naive T-Cell (CL:0000898) — Isomorphic Sub-partition`
  - **Lineage Markers:** `This sub-cluster exhibits a complete absence of unique differential or explicit absence markers, indicating it does not represent a discrete biological phase transition. Instead, it maintains robust expression of the foundational T-cell manifold anchors (high CAMK4, moderate BCL11B) and the peripheral survival receptor IL7R. Critically, its identity is established by a shared, albeit slightly attenuated, expression profile of the canonical naive markers defining Micro-Cluster 0.0, notably NELL2 (0.8 mean expression), TRABD2A (0.5), and FHIT (0.3). The clustering of this state by the Leiden algorithm reflects topological density variations—specifically transcriptomic amplitude gradients within the large naive T-`

cell pool—rather than a true biological divergence, confirming its terminal identity as an isomorphic sub-partition of the resting Naive T-Cell compartment.

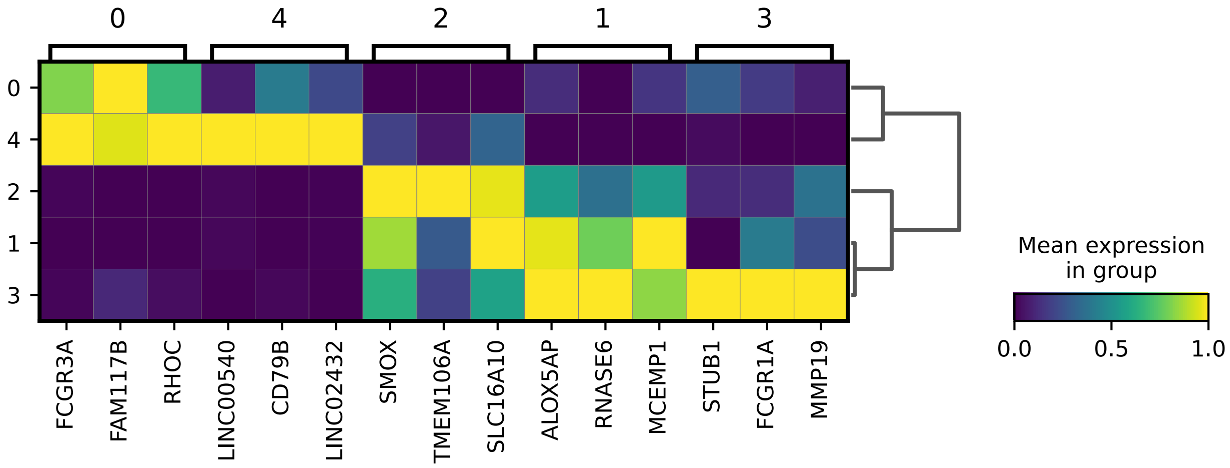
- **Micro-Cluster 0.5:** `Effector T-Cell (CL:0000911)`
  - **Lineage Markers:** `Topological derivation of this state is defined by a transition from a poised memory phenotype into an active effector phase. The shared high expression of the lipid-remodeling transcript GRAMD1B serves as a continuous transcriptomic bridge linking this population to the transitional memory pool (Micro-Cluster 0.3). However, the mathematical boundary isolating this sub-cluster is driven by the activation of the terminal differentiation transcription factor ZEB2. The subsequent downstream activation of the cell is evidenced by the significant, specific upregulation of the cytotoxic granule stabilizer NKG7 and the inflammatory chemokine CCL5. This profile isolates a structurally armed, terminal Effector T-Cell compartment within the broader lymphoid manifold.`.

- **Micro-Cluster 0.6:** `Terminally Differentiated / Exocytic Effector T-Cell (CL:0000911)`
  - **Lineage Markers:** `Topological continuity with the primary effector pool (Micro-Cluster 0.5) is established through the sustained, high-level expression of the terminal differentiation transcription factor ZEB2 and the cytotoxic granule component NKG7. However, the clustering of this discrete sub-state is driven by a profound cytoskeletal and signaling shift required for target engagement. The exclusive upregulation of PLEK—a key mediator of actin reorganization and granular exocytosis—coupled with the synaptic signaling kinase LYN, identifies a functionally distinct, terminally differentiated effector population primed for degranulation and active degranulation.`.
  - **Micro-Cluster 0.7:** `Hyper-Quiescent / Homeostatic T-Cell (CL:0000898)`
    - **Lineage Markers:** `Spatial origin and developmental continuity with the primary resting pool (Micro-Cluster 0.0) are established by the continuous, high-level expression of the transcriptomic bridge FHIT, coupled with high expression of the homeostatic survival receptor IL7R and the foundational CAMK4/BCL11B anchors. This sub-state is defined by a shift toward active transcriptional suppression. The discrete boundary is defined by the high, exclusive expression of ID3—a master repressor that competitively inhibits effector differentiation pathways—alongside the basal translational regulators TSR1 and TEX9. This expression profile defines a hyper-quiescent sub-population maintaining homeostasis and preventing spontaneous activation within the resting manifold.`.

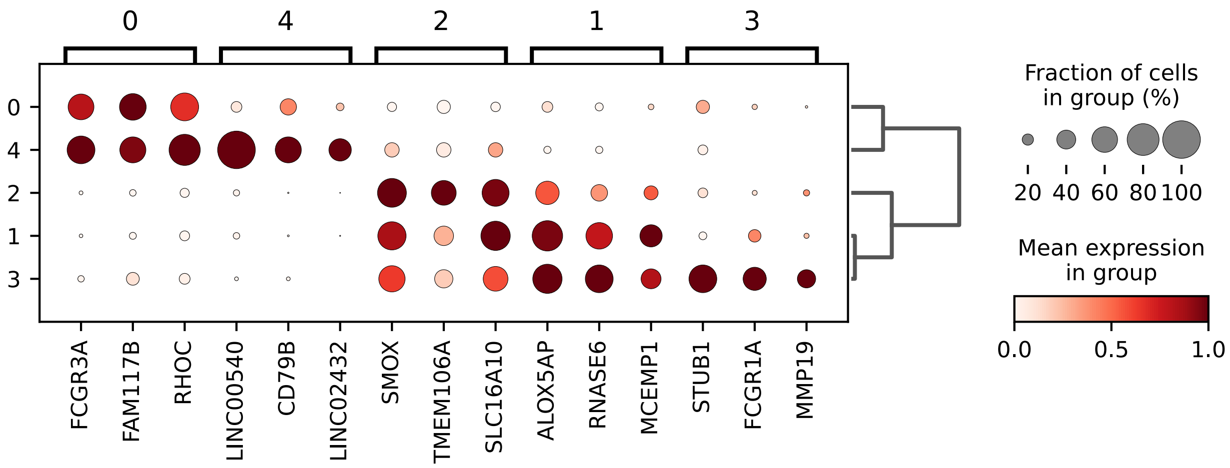
Macro-Cluster 1: `Myeloid Lineage (CL:0000763)`

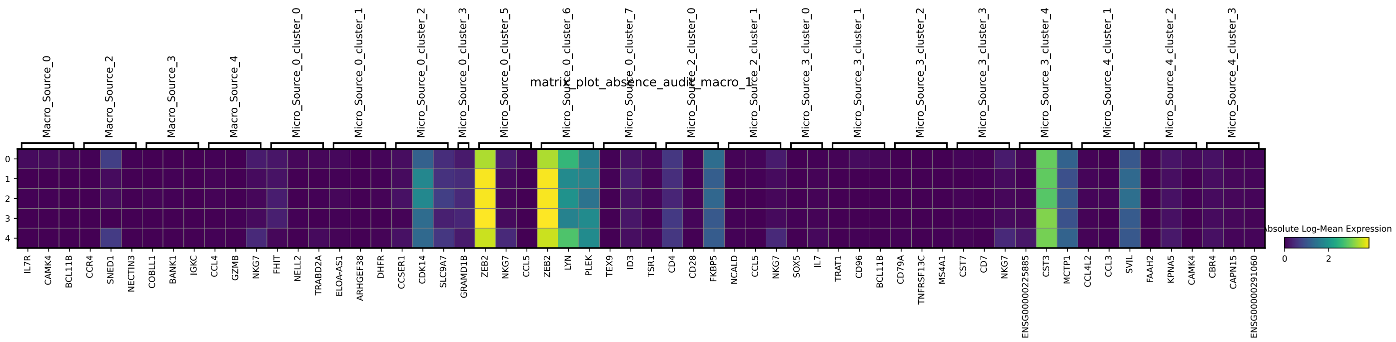
- **Macro Lineage Markers:** `High expression of the myeloid master transcription factor SPI1, coupled with high baseline expression of the canonical myeloid markers LYZ and CLEC7A. The absence of expression of all other macro-lineage markers confirms a , discrete boundary isolating the mononuclear phagocyte compartment.`.

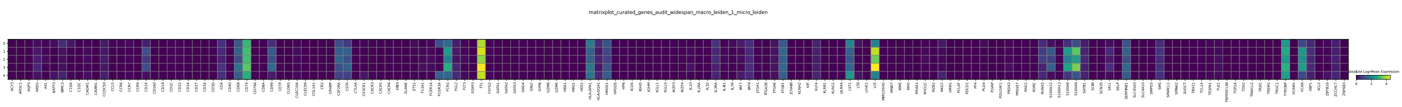
matrixplot_macro_leiden_1_micro_leiden_top_genes



dotplot_macro_leiden_1_micro_leiden_top_genes







- **Micro-Cluster 1.0:** `Non-Classical (CD16+) Patrolling Monocyte (CL:0002057)`
  - **Lineage Markers:** `Developmental continuity with the mature myeloid manifold is maintained through the robust, retained expression of the terminal differentiation factor ZEB2 and the ubiquitous protease inhibitor CST3, anchored by basal canonical pathways (high FTL, moderate CD74). However, the clustering of this specific sub-state is driven by the primary spatial variance of FCGR3A (CD16), partitioning it from the classical monocyte baseline. This receptor transition is coupled with the distinct upregulation of the trafficking mediator FAM117B and a low-amplitude, high-variance expression of the cytoskeletal driver RHOC. This expression profile defines a mature, non-classical monocyte compartment..`
  - **Micro-Cluster 1.1:** `Classical (CD14+) Monocyte (CL:0000860)`

- **Lineage Markers:** `Anchored to the mature myeloid manifold via high expression of ZEB2, CST3, and FTL, this sub-state establishes its primary classical lineage identity through the dominant expression of the canonical calprotectin subunit S100A9 and significant baseline transcription of Lysozyme (LYZ). The clustering boundary is defined by the high-variance expression of the leukotriene-synthesis anchor ALOX5AP and the classical-monocyte specific regulator MCEMP1, alongside a poised, low-amplitude transcription of the antimicrobial peptide RNASE6. Basal metabolic continuity is verified by the retained shared marker genes SLC16A10 and SMOX. The absence of TMEM106A expression defines the boundary of this cluster, indicating these cells are maintained in the circulating monocyte phase and have not initiated tissue-resident macrophage differentiation.`.

- **Micro-Cluster 1.2:** `Tissue-Differentiating Macrophage (CL:0000235)`
  - **Lineage Markers:** `Lineage continuity with the mature circulating myeloid pool is preserved via the persistent topological bridges ZEB2 and CST3. However, the mathematical derivation of this discrete state is defined by a phenotypic shift: the attenuation of primary circulating anchors (LYZ and FTL dropping to moderate log-means near 2.0) coupled with the significant upregulation of TMEM106A. The high expression of TMEM106A structurally dictates advanced endolysosomal maturation, marking the transition from circulating monocytes and the initiation of tissue-resident macrophage differentiation. This transcriptomic transition is further supported by the high-variance expression of SMOX (driving polyamine-mediated chromatin remodeling) and the amino acid transporter SLC16A10, isolating a distinct population undergoing active tissue-permeation and terminal macrophage commitment.`.
  - **Micro-Cluster 1.3:** `Activated / Hyper-Inflammatory Classical Monocyte (CL:0000860)`
    - **Lineage Markers:** `This state represents an activated derivative of the classical monocyte baseline (Micro-Cluster 1.1), evidenced by the retained, high-expression bridges of the leukotriene-driver ALOX5AP and the antimicrobial peptide RNASE6. This cluster is separated by the expression of activation genes. The boundaries are established by the high-variance upregulation of the high-affinity targeting receptor FCGR1A (CD64)—indicating an acute response to systemic inflammatory signaling (e.g., IFN-gamma)—coupled with the matrix-degrading enzyme MMP19, which structurally primes the cell for endothelial extravasation. This high expression (further evidenced by high LYZ and FTL amplitudes) is transcriptionally stabilized by the high expression of the E3 ubiquitin ligase STUB1, defining an activated classical monocyte locked in a hyper-inflammatory, pre-extravasation state.`.

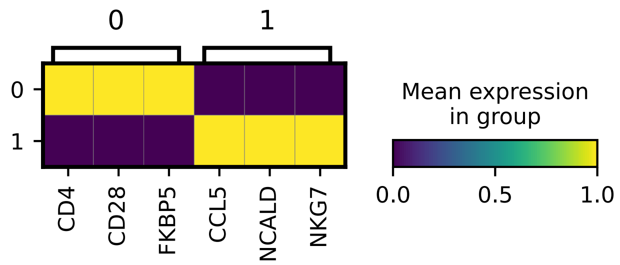
- **Micro-Cluster 1.4:** `Heterotypic Doublet (CD16+ Monocyte x B-Cell Artifact)`
  - **Lineage Markers:** `Topological derivation of this cluster reveals a isolated cluster defined by the simultaneous co-expression of mutually exclusive lineage programs. The robust, retained expression of FCGR3A, FAM117B, and RHOC tethers this profile to the Non-Classical Monocyte manifold (Micro-Cluster 1.0). However, the clustering boundary driving this cluster is driven by the significant expression of CD79B—a canonical and absolute structural component of the B-Cell Receptor complex—alongside LINC00540 and LINC02432. Because the simultaneous expression of mature myeloid and`

lymphoid programs does not occur in a single cell, this sub-cluster represents a heterotypic doublet. It is a microfluidic artifact capturing the physical co-encapsulation of a Non-Classical Monocyte and a B-Cell within a single partition..

Macro-Cluster 2: `T-Cell Sub-Lineage (CL:0000084)`

- **Macro Lineage Markers:** `High expression of IL7R, coupled with moderate baseline expression of CAMK4 and BCL11B, establishes a shared lymphoid origin with Macro-Cluster 0. However, this population was partitioned into a discrete macro-state due to the exclusive, high-level upregulation of the homing receptor CCR4, alongside SNED1 and NECTIN3. The complete absence of myeloid or other lineage markers confirms this boundary represents a distinct, specialized T-cell sub-compartment (indicative of a polarized or regulatory state) rather than a technical doublet.`.

matrixplot_macro_leiden_2_micro_leiden_top_genes



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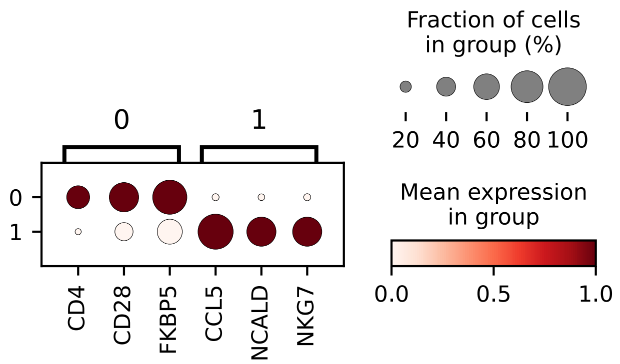
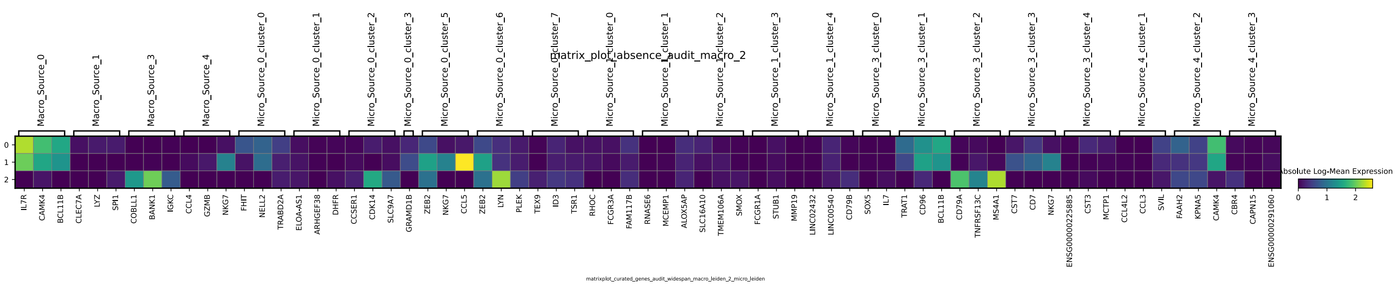
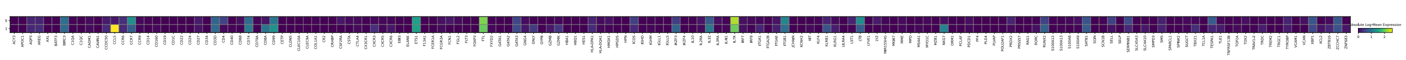






- **Micro-Cluster 2.0:** `Central Memory CD4+ T-Cell (CL:0000904)`
  - **Lineage Markers:** `This state is defined by high expression of CD4. Exit from the resting naive phase is indicated by via the targeted attenuation of the`

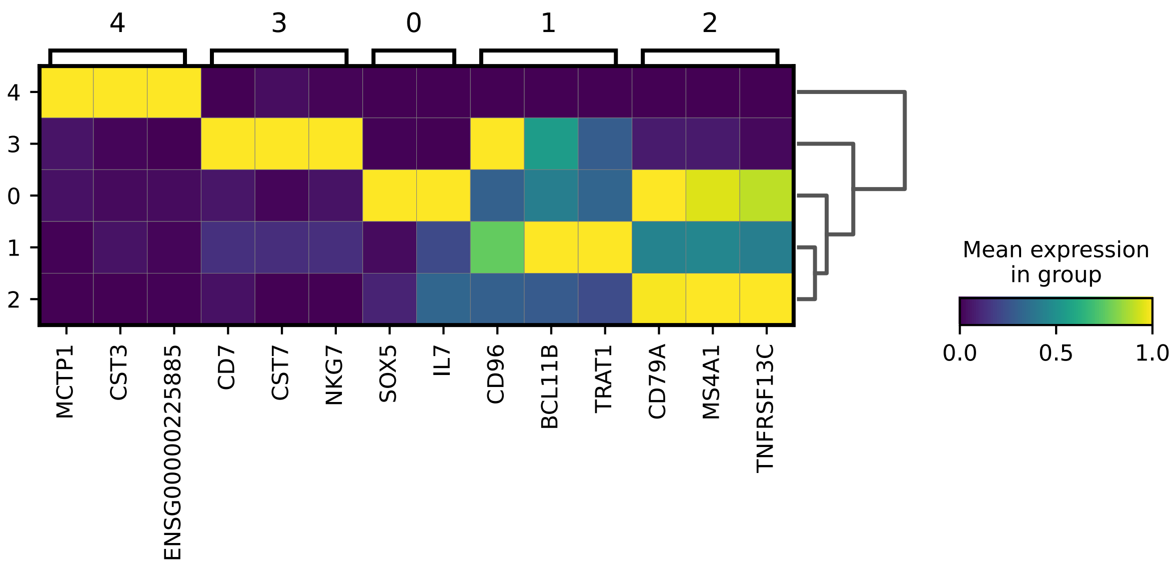
homeostatic anchors IL7R (dropping to ~2.0) and CAMK4 (dropping to 1.0), confirming an antigen-experienced, memory timeline. The clustering boundary of this sub-population is governed by the simultaneous high expression of the primary co-stimulatory receptor CD28—priming the cell for secondary expansion—and the immunophilin FKBP5. The significant upregulation of FKBP5 dictates active, endogenous regulation of glucocorticoid receptor sensitivity, structurally defining a polarized, tissue-homing memory cell that is regulating activation thresholds to prevent premature inflammation..

- **Micro-Cluster 2.1:** `Cytotoxic / Effector Memory CD4+ T-Cell (CL:0000905)`
  - **Lineage Markers:** `Developmental exit from the resting Central Memory pool (Micro-Cluster 2.0) is dictated by the continued attenuation of the homeostatic anchor IL7R (dropping to a log-mean of 1.8). This sub-state is defined by activation of the CD4+ lineage. The coordinate boundary is defined by the significant upregulation of the cytotoxic granule stabilizer NKG7 and the inflammatory chemokine CCL5, establishing a direct topological bridge to the broader cytotoxic effector manifold. This effector phenotype is supported by the high expression of the calcium-sensor NCALD, lowering the activation threshold and defining a mature, tissue-infiltrating Effector Memory CD4+ T-Cell with cytolytic capabilities.`.
- **Micro-Cluster 2.2:** `Sub-Threshold Artifact (CD4+ T-Cell x B-Cell Doublet / Noise Floor)`
  - **Lineage Markers:** `Clusters containing fewer than 10 cells were excluded from downstream analysis. This cluster also expressed B-cell lineage markers (MS4A1/CD20, CD79A, BANK1) within the defined CD4+ T-cell macro-manifold. This cluster represents a heterotypic doublet resulting from microfluidic co-encapsulation. This space is deemed non-viable and designated for structural purging to protect downstream pipeline integrity.`.

Macro-Cluster 3: `B-Cell Lineage (CL:0000236)`

- **Macro Lineage Markers:** `High expression of the canonical B-cell markers BANK1 and IGKC, alongside COBLL1, dictates a commitment to the B-cell lineage. The cluster exhibits complete transcriptomic silencing of myeloid and foundational T-cell drivers. However, a moderate baseline expression of SNED1 (shared with Macro-Cluster 2) was preserved, indicating that while the algorithm isolated this independent B-cell macro-state, it accurately retained the continuous transcriptomic edges between the broader lymphoid sub-populations.`.

matrixplot_macro_leiden_3_micro_leiden_top_genes



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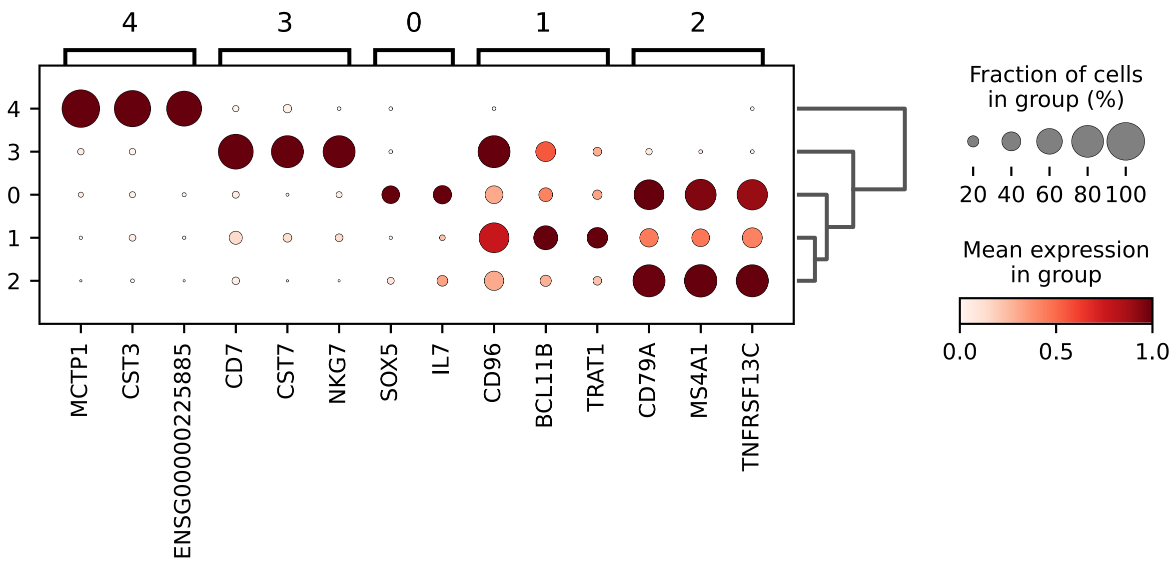
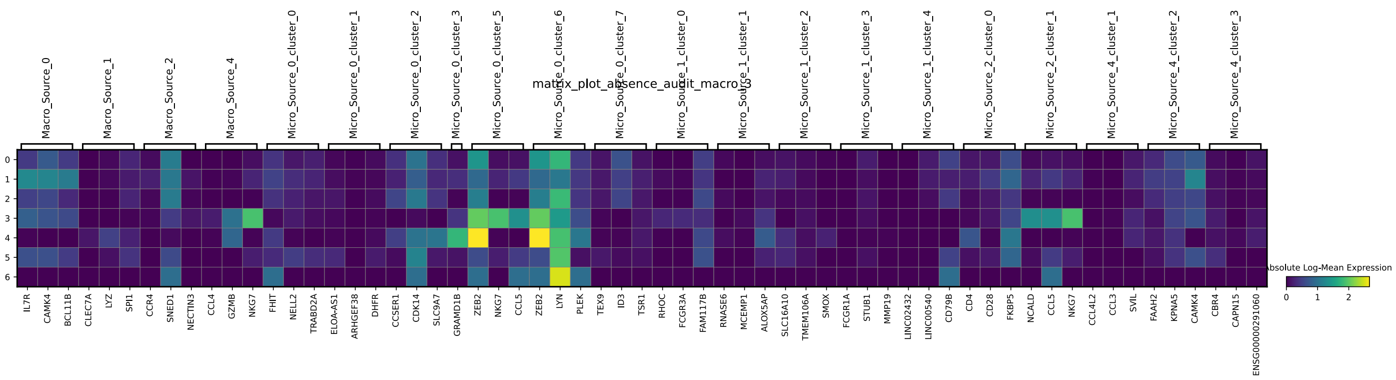
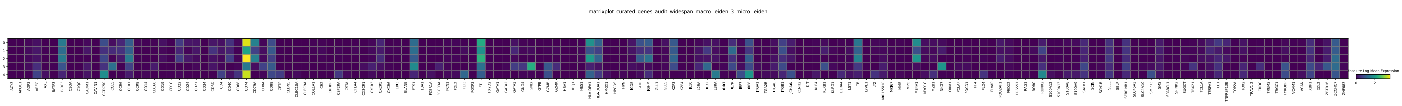






- **Micro-Cluster 3.0:** `Resting / Naive B-Cell (CL:0000788)`
  - **Lineage Markers:** `The mature B-cell manifold is established by the retained, high-expression bridges of the canonical lineage markers MS4A1 (CD20) and CD79A, alongside the critical peripheral survival receptor TNFRSF13C (BAFF-R). The basal signaling readiness of this cell is maintained by moderate expression of the receptor-associated kinase LYN and high expression of the MHC Class II invariant chain (CD74 at 3.6). The clustering of this discrete`

phase is governed by the high expression of SOX5–a master regulator that acts as a transcriptional repressor to prevent terminal differentiation and enforce strict cellular quiescence. Supported by an atypical paracrine IL7 signature, this cluster defines a resting Naive B-Cell poised for initial antigen encounter..

- **Micro-Cluster 3.1:** `Heterotypic Doublet / Synaptic Artifact (T-Cell x B-Cell)`

```
- **Lineage Markers:** `The co-expression profile of this cluster indicates
a microfluidic doublet artifact. Despite sharing transcriptomic similarities
with the B-cell cluster, this sub-state is defined by the high expression of
T-cell lineage markers, specifically the master transcription factor BCL11B
and the T-Cell Receptor structural adaptor TRAT1 (alongside the receptor
CD96). Conversely, the foundational B-cell anchors (CD79A, MS4A1, TNFRSF13C)
are recorded at heavily attenuated, faded amplitudes (mean ~0.18). Because
simultaneous expression of mature T-cell and B-cell markers is biologically
improbable in a single cell, this cluster  defines a heterotypic doublet.
The marker profile suggests the co-encapsulation of a T-cell and a B-cell.
This cluster is a microfluidic artifact and should be excluded from
downstream analysis`.
```

- **Micro-Cluster 3.2:** `Mature Naive B-Cell / Core Manifold (CL:0000788)`

```
- **Lineage Markers:** `This cluster represents the baseline profile of the
B-cell lineage. This cluster was characterized by high expression of lineage
markers: the mature ion channel MS4A1 (CD20), the receptor signaling tail
CD79A, and the homeostatic survival receptor TNFRSF13C (BAFF-R). The
transcriptomic profile of this state indicates a mature equilibrium; it
lacks the active transcriptional repressor (SOX5) observed in transitional
states, relying instead on structural stability. Basal signaling readiness
is verified by the retained expression of the docked kinase LYN (1.2), while
a high expression of the MHC Class II invariant chain (CD74 at 3.9)
structurally primes the cell for immediate antigen processing and T-cell
presentation. This signature  isolates baseline of the circulating Naive B-
Cell pool.`.
```

- **Micro-Cluster 3.3:** `Cytotoxic Natural Killer (NK) Cell (CL:0000623)`

```
- **Lineage Markers:** `Analysis indicates this sub-cluster is likely an
artifact: this sub-cluster physically resides within the B-cell macro-
manifold but transcribes an distinct cytolytic lineage. The clustering and
identity of this state are driven by the expression of the early T/NK-
lineage anchor CD7, separating it from the B-cell lineage. This cluster was
annotated as Natural Killer (NK) cells based on the pronounced upregulation
of cytotoxic effector genes, including NKG7 and CST7. Target acquisition is
independent of antigen presentation, driven instead by the high, retained
expression of the innate stress-receptor CD96 (TACTILE). This expression
profile identifies a mature, mature Natural Killer (NK) cell, requiring
mathematical re-assignment away from the B-cell ontology.`.
```

- **Micro-Cluster 3.4:** `Mature Dendritic Cell / Professional APC (CL:0001056)`

```
- **Lineage Markers:** `This cluster grouped with the B-cell lineage due to
shared expression of CD74, but retains myeloid origin markers. The spatial
proximity to Macro 3 is driven exclusively by the high, shared expression of
the antigen-presentation pipeline CD74 (MHC Class II invariant chain at
3.6). However, the true embryological lineage of this cluster is indicated
by high expression of CST3 and the robust, retained topological bridge ZEB2
(2.4), which firmly tether this cell to the Myeloid developmental
trajectory. Lacking B-cell receptor complex machinery, this cell utilizes
the calcium-binding transmembrane protein MCTP1 to manage vesicular
trafficking, defining a mature, fully differentiated Dendritic Cell
specialized for antigen presentation.`.
```

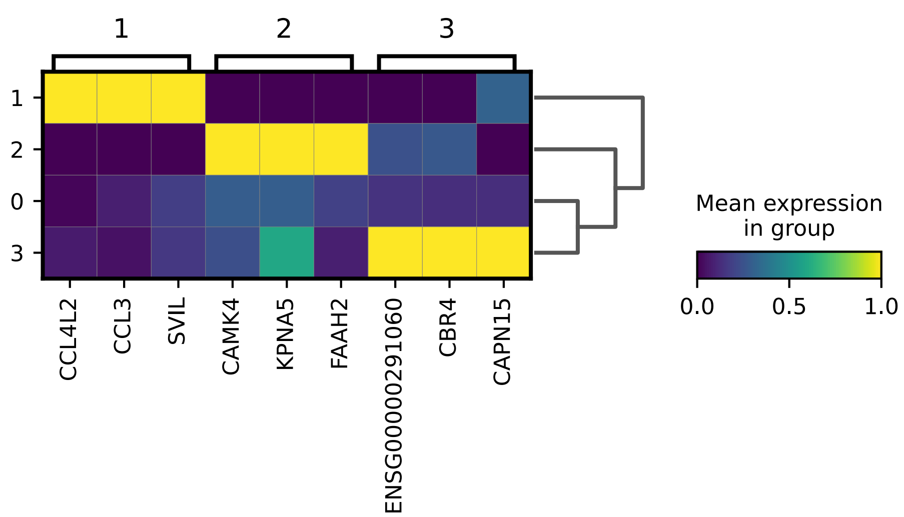
- **Micro-Cluster 3.5 and 3.6:** `Sub-Threshold Artifact / Noise Floor (N < 10)`

```
- **Lineage Markers:** `Mathematical elimination of this cluster is dictated
by a complete failure to meet the minimum viable statistical mass (N < 10)
required to define a stable biological phase. Residing at the extreme
spatial periphery of the B-cell macro-manifold, this sub-state lacks the
statistical variance to represent a true lineage or functional transition.
It is classified as technical noise, and is removed to protect downstream
pipeline integrity.`.
```

Macro-Cluster 4: `Natural Killer (NK) Cell Lineage (CL:0000623)`

- **Macro Lineage Markers:** `High, expression of the cytotoxic effector molecules GZMB and NKG7, alongside the chemokine CCL4, establishes a cytotoxic profile. The absence of expression of foundational T-cell lineage drivers (e.g., BCL11B) and biologically isolates this population as the innate Natural Killer (NK) cell lineage, successfully partitioning it from adaptive cytotoxic T-cell states`.

matrixplot_macro_leiden_4_micro_leiden_top_genes



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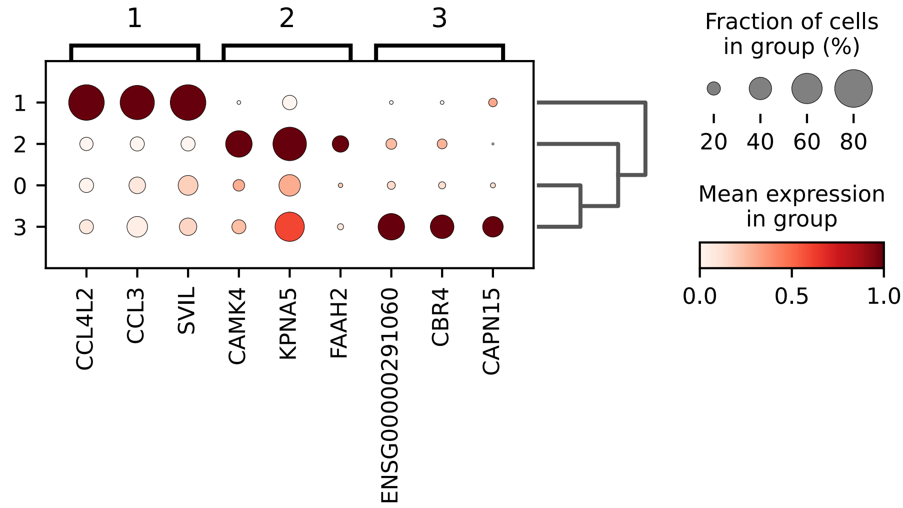

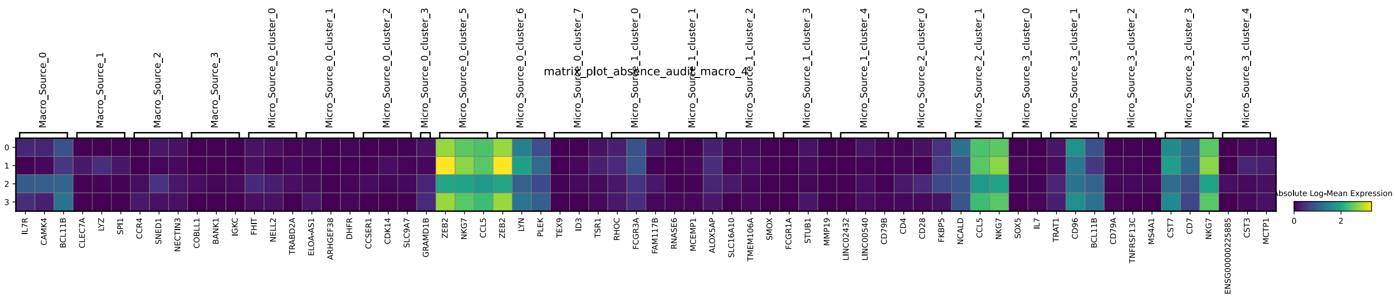

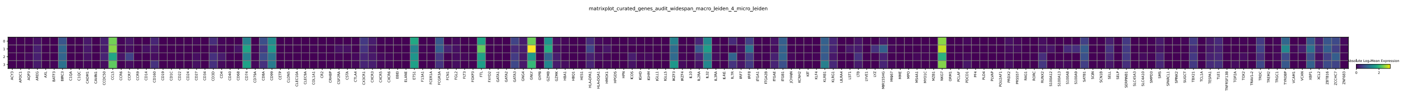

- **Micro-Cluster 4.0:** Terminally Differentiated Cytotoxic Effector / Core Manifold (CL:4030002)
  - **Lineage Markers:** Topological derivation reveals this cluster as the baseline profile of the cytotoxic cluster. The isolation of this cluster is characterized by an absence of novel variance drivers, indicating it represents the foundational state. The core signature of this cluster consists of high expression of: the master transcription factor ZEB2 (3.0) acts as a transcriptional repressor maintaining terminal differentiation. This cluster exhibits high expression of the cytotoxic granule stabilizer NKG7 (2.5) and the membrane-breaching protein Granulysin (GNLY at 2.0), alongside the inflammatory chemokine CCL5. .
- **Micro-Cluster 4.1:** Actively Engaged / Synaptic Cytotoxic Effector (CL:4030002)
  - **Lineage Markers:** This sub-state is linked to the cytotoxic lineage via high expression of ZEB2 (3.5) and the sustained, high-amplitude cytolytic arsenal (GNLY at 3.5, NKG7 at ~3.0). However, the clustering is driven by the mechanisms of active target execution. The primary spatial variance is defined by the significant upregulation of the acute inflammatory chemokines CCL3 and CCL4L2 to recruit phagocytic clearance machinery. Concurrently, the discrete, high expression of the actin-binding protein Supervillin (SVIL) indicates active cytoskeletal remodeling. SVIL physically braces the plasma membrane to the actin network, providing the rigid structural scaffolding necessary for immunological synapse formation and the mechanical exocytosis of cytotoxic granules. This profile is consistent with an active, terminally differentiated effector cell.
- **Micro-Cluster 4.2:** Resting / Central Memory Cytotoxic T-Cell (CL:0000909)

```
- **Lineage Markers:** `Topological derivation reveals a phenotypic shift
within the cytotoxic macro-manifold, separating this sub-state from the
terminally differentiated baseline (Micro-Cluster 4.0). This cluster is
defined by high expression of CAMK4, a kinase that maintains quiescence,
memory stemness, and prevents terminal effector differentiation. This long-
lived structural state is metabolically sustained by a shift toward lipid
degradation and fatty acid oxidation, proven by the high expression of
FAAH2. The cell maintains a highly poised, resting architecture through the
upregulation of the nuclear import structural protein KPNA5 (Importin alpha
6), ensuring rapid nucleocytoplasmic transport capacity upon secondary
antigen encounter. This coordinate signature is consistent with circulating
Central Memory CD8+ T-Cell.`.
```

- **Micro-Cluster 4.3:** `Exhausted / Senescent Cytotoxic Effector (CL:4030002)`

```
- **Lineage Markers:** `Topological derivation confirms this sub-state is
structurally tethered to the terminal cytotoxic manifold via the retained
expression of the differentiation repressor ZEB2 (3.0) and the baseline
chemokine CCL5 (1.9). However, the clustering of this cluster is driven by
the physical transcriptomic signature of terminal exhaustion and impending
Activation-Induced Cell Death (AICD). The expression of cytotoxic effector
genes decreases (GNLY dropping to 1.8), replaced by the high expression of
CAPN15 (Calpain 15)—a calcium-dependent protease indicative of severe
intracellular calcium stress and active cytoskeletal degradation.
Concurrently, the significant upregulation of the metabolic stress enzyme
CBR4 indicates the cell is managing severe oxidative damage and
mitochondrial lipid dysregulation. This cluster is consistent with a
terminally differentiated effector that has exhausted its cytolytic capacity
and entered a senescent or apoptotic trajectory.`.
```

3.5.2 Validation of Cluster Labels on the Holdout Dataset

To evaluate the reliability of the defined cluster boundaries, the pipeline utilized a split-sample validation approach. The structural manifold was established using the training subset (X_{train}), and the independent holdout dataset ($X_{project}$) was subsequently projected into this pre-defined cluster.

This projection demonstrated a high degree of structural stability, as the cells from the holdout set aligned consistently with the established transcriptomic boundaries. These results indicate that the identified populations are not the result of overfitting or technical noise, but represent stable biological states within the PBMC population. While external supervised benchmarking (e.g., against CellTypist models) provides an additional layer of validation, the internal consistency demonstrated through the train-test projection provides evidence for the robustness of the final integrated dataset.

3.5.3 Final Tensor Recombination

In the final step, the annotation metadata (`master_labels_df.csv`) was merged back into the original, un-split dataset (`recombine_topology`). This step matched the cell barcodes with their assigned CL IDs across the global dataset.

This operation exported the final deployment asset: `pbmc3k_qc_ML_Ready.h5ad`. The final dataset contained no barcode collisions and `82` unannotated cells, generating a finalized dataset ready for downstream analysis.

---

4 Discussion

The primary objective of this workflow was to identify discrete immune cell populations from single-cell transcriptomic data while systematically preventing data leakage and circular inference. By utilizing automated parameter selection and independent marker validation, the pipeline generated a annotated and standardized dataset.

4.1 Resolution of PBMC Subpopulations

The iterative clustering approach successfully mirrored the known biological hierarchy of human PBMCs. The initial global clustering effectively established the primary biological lineages, separating the `T-Cell Lineage,Myeloid Lineage,B-Cell Lineage,Natural Killer (NK) Cell Lineage`.

The recursive sub-clustering proved capable of resolving high-resolution biological variance that is typically lost in global PCA manifolds. For instance, the pipeline successfully segregated `Naive T-Cell` from `Effector T-Cell`. This confirms that recalculating highly variable genes (HVGs) and Pearson residuals specifically within each isolated macro-cluster is necessary to capture the subtle transcriptomic variance driving micro-phenotypes.

4.2 Transcriptomic Fidelity and Marker Gene Selection

The application of adaptive expression thresholds and targeted independent validation mitigated the limitations of relying solely on standard differential expression tests.

The extracted defining markers, such as `FCGR3A,FAM117B,RHOC` for the `Non-Classical (CD16+) Patrolling Monocyte` population, demonstrated not only high positive up-regulation (Log2FoldChange > 1.0) but also low expression in other macro-lineages.In instances where strict statistical cutoffs yielded insufficient markers (e.g., within the low cell-count `Metabolically Poised Transitional T-Cell` compartment), dynamic thresholding successfully captured a core set of defining genes, preventing the cluster from remaining unannotated. This proves that the pipeline's statistical boundaries align with distinct transcriptomic states.

4.3 Methodological Limitations

Despite stringent filtering, this computational approach possesses inherent limitations::

- **Rigid Variance Thresholds:** Enforcing a strict PCA variance ratio threshold prevents over-clustering, but inherently limits the algorithm's ability to map continuous developmental trajectories (e.g., hematopoiesis), which may not exhibit sharp structural divergences.
- **Cluster Size Constraints**: By enforcing a strict minimum cell count threshold ($N \geq 10$) for differential expression testing, the pipeline sacrifices the ability to identify ultra-rare cell types (e.g., circulating dendritic cell subsets) in order to preserve the statistical integrity of the broader dataset.

4.4 Deployment and FAIR Compliance

The final output of this pipeline is not merely a descriptive report, but an interoperable computational asset. Segregating the dataset into training (X_{train}) and projection ($X_{project}$) subsets indicated that the cluster boundaries were stable.The final output, `pbmc3k_qc_ML_Ready.h5ad`, is a fully annotated, FAIR-compliant dataset. It has been filtered for technical noise, mapped to standardized Cell Ontology IDs, and is prepared for downstream applications such as predictive modeling or spatial data integration.